



Fundamentals of neurobiology

ch5 : plasticity

R.C. Lambert

Neuronal networks and physiopathological rhythms

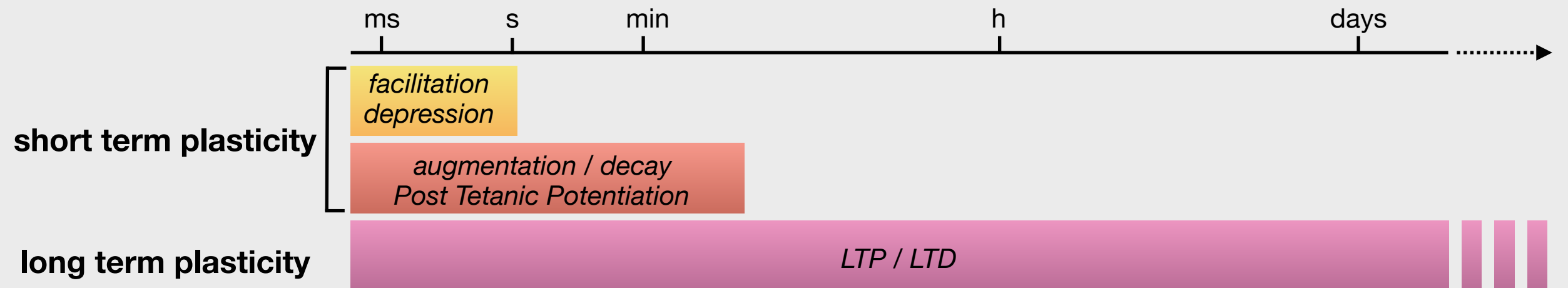
<http://www.ibps.upmc.fr/fr/Recherche/umr-8246/reseaux-de-neurones-et-rythmes-physiopathologiques>

Sorbonne Université- Neuroscience Paris Seine (NPS)

CNRS UMR8246, INSERM U1130, UPMC UMI19



different time scales of synaptic plasticity

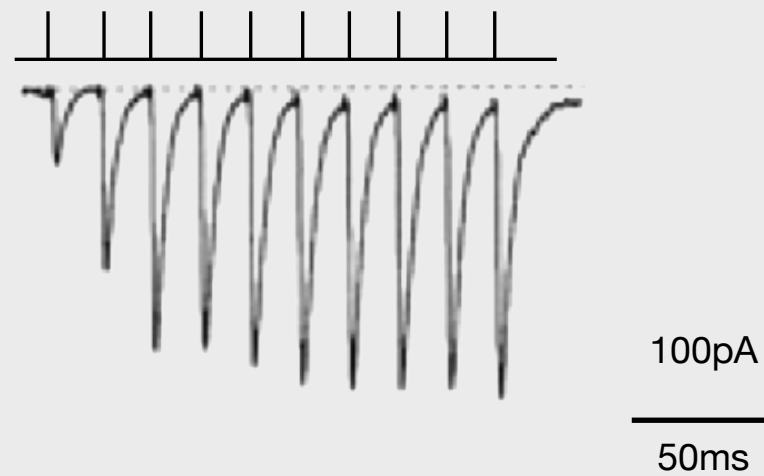


short term plasticity: facilitation and depression

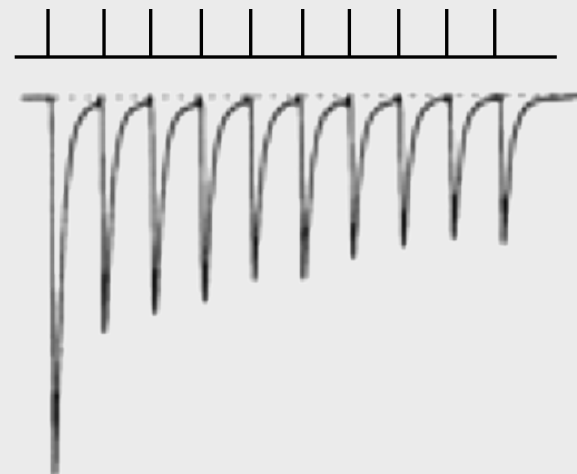
excitatory post-synaptic current (EPSC)



parallel fiber stimulation (75Hz)



climbing fiber stimulation (75Hz)

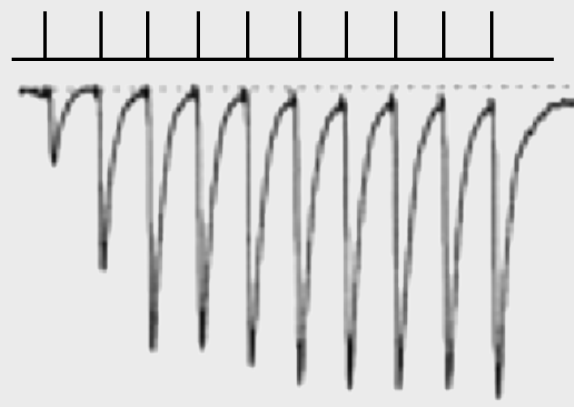


short term plasticity: facilitation and depression

excitatory post-synaptic current (EPSC)



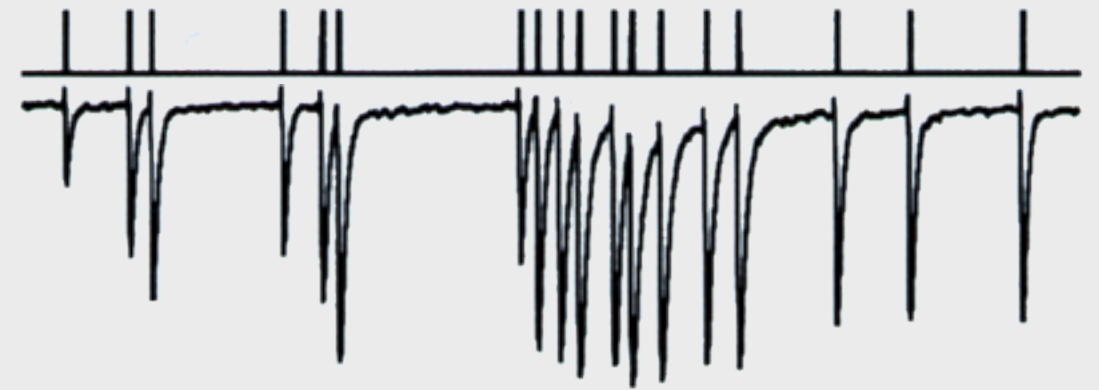
parallel fiber stimulation (75Hz)



100pA

50ms

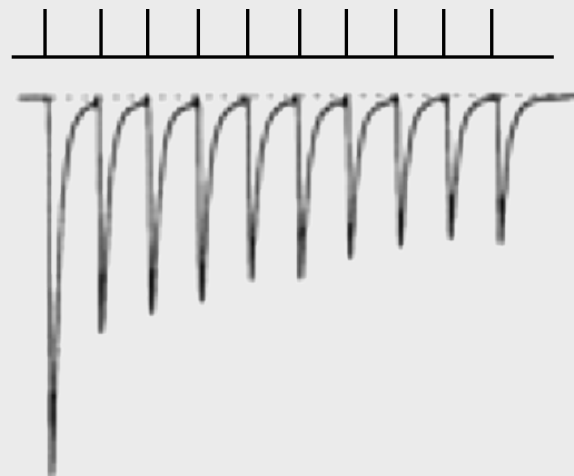
natural pattern



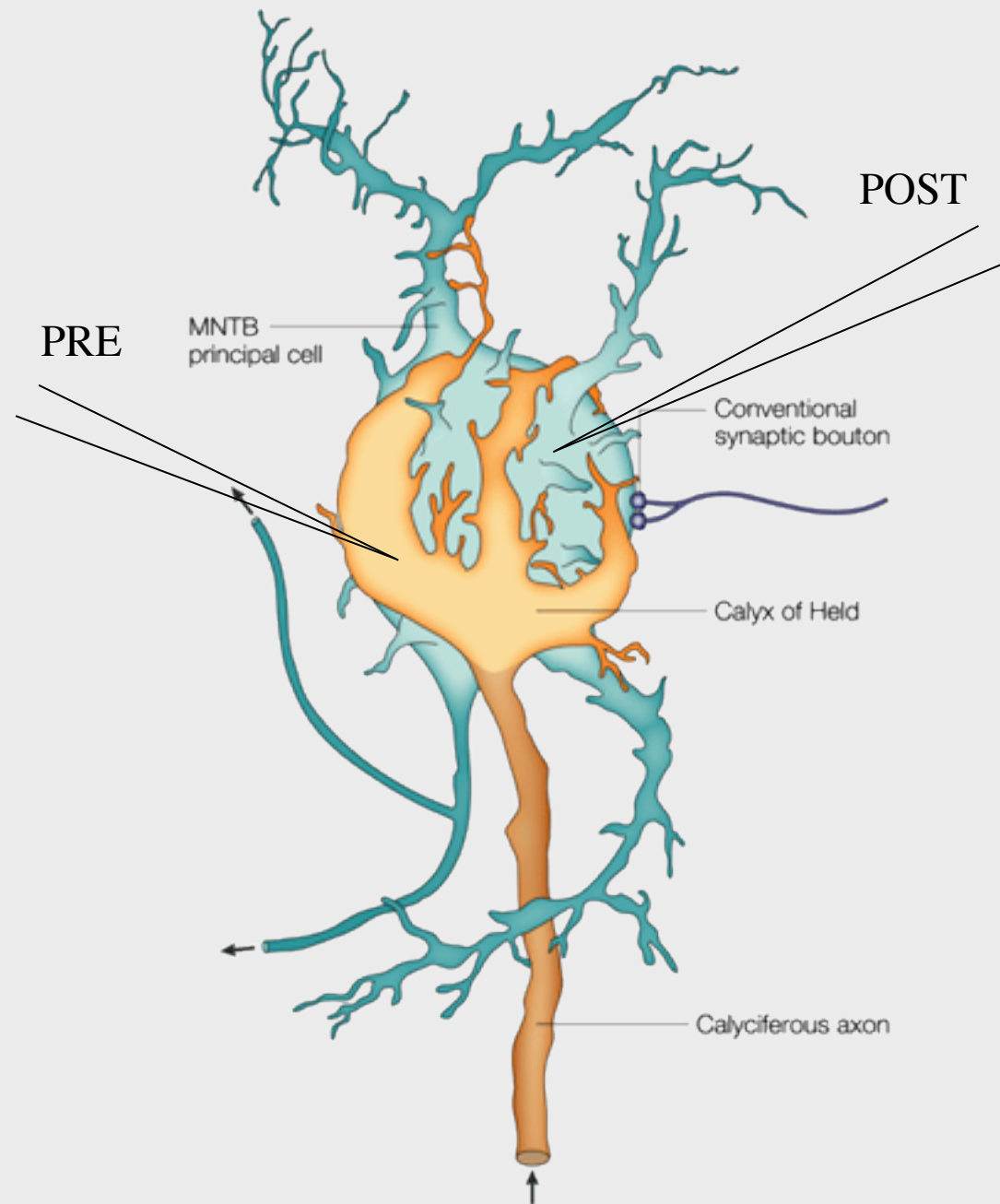
500pA

200ms

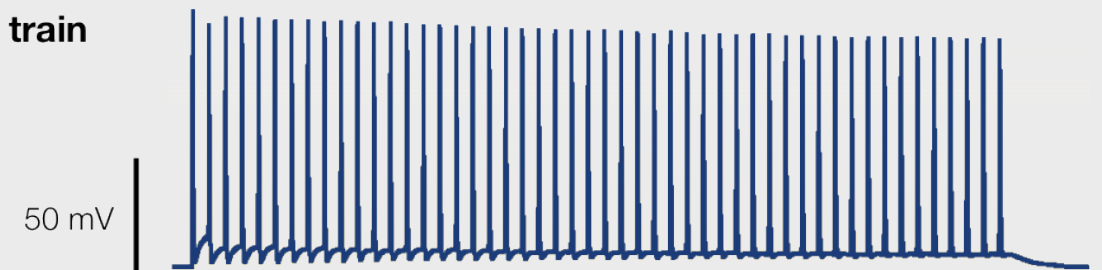
climbing fiber stimulation (75Hz)



short term plasticity: facilitation and depression



Presynaptic AP train
(100 Hz)



EPSC
(external $[Ca^{2+}] = 1 \text{ mM}$)

low probability

200 pA
50 ms



EPSC
(external $[Ca^{2+}] = 2 \text{ mM}$)

high probability



Borst et al., 1995

low probability of release : facilitation

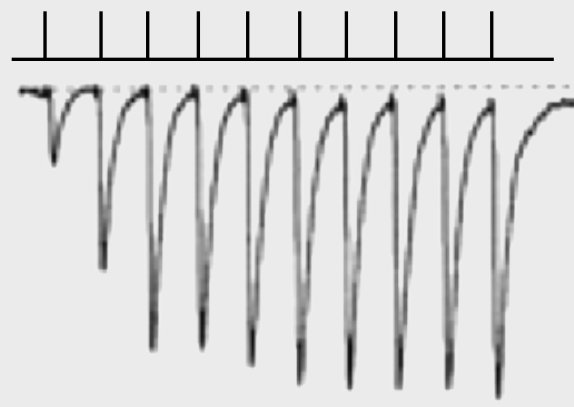
high probability of release : depression

short term plasticity: facilitation and depression

excitatory post-synaptic current (EPSC)



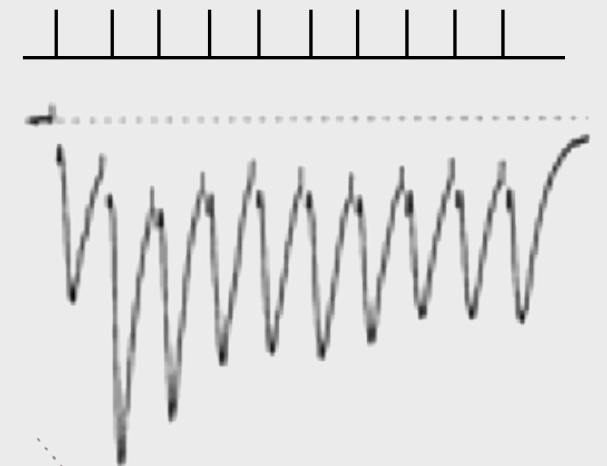
parallel fiber stimulation (75Hz)



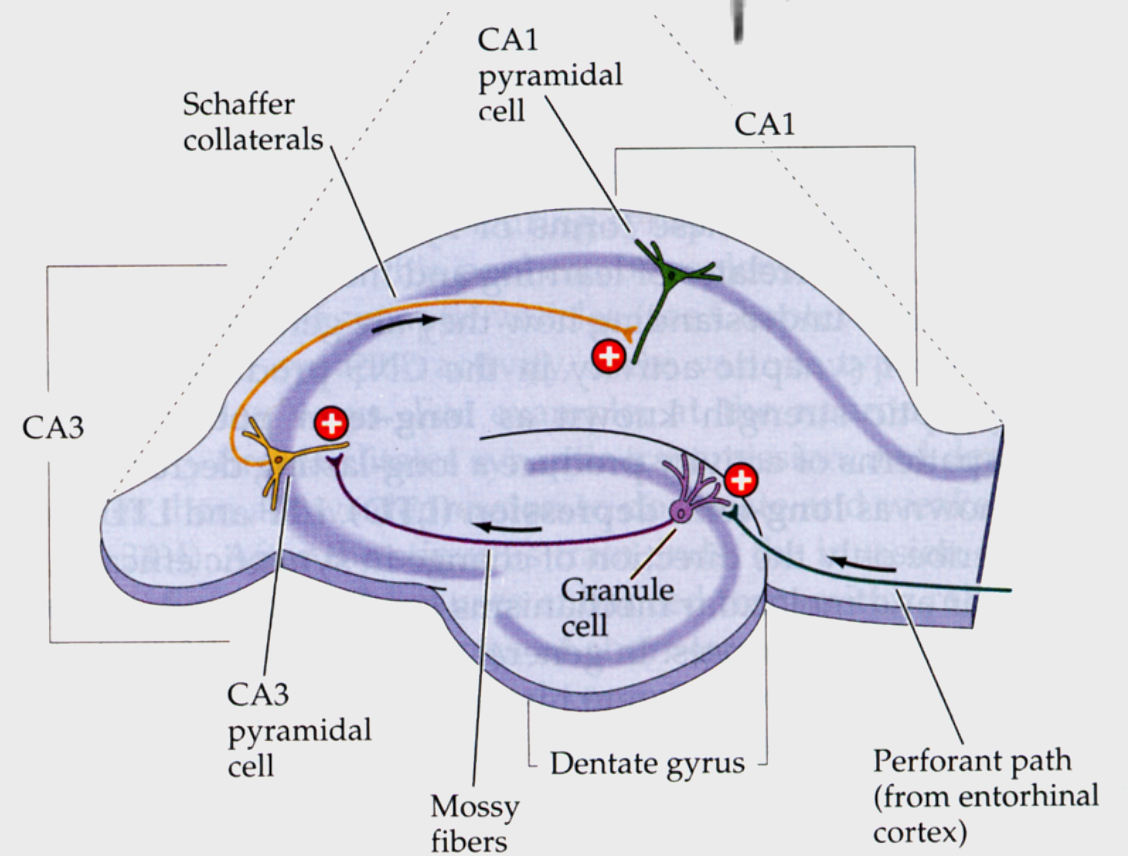
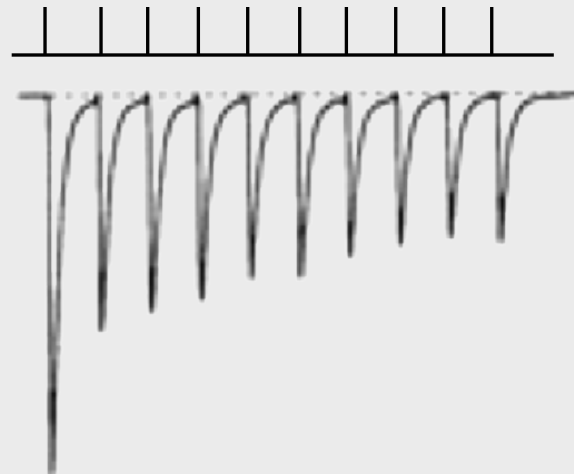
100pA

50ms

Schaffer collateral stimulation (75Hz)

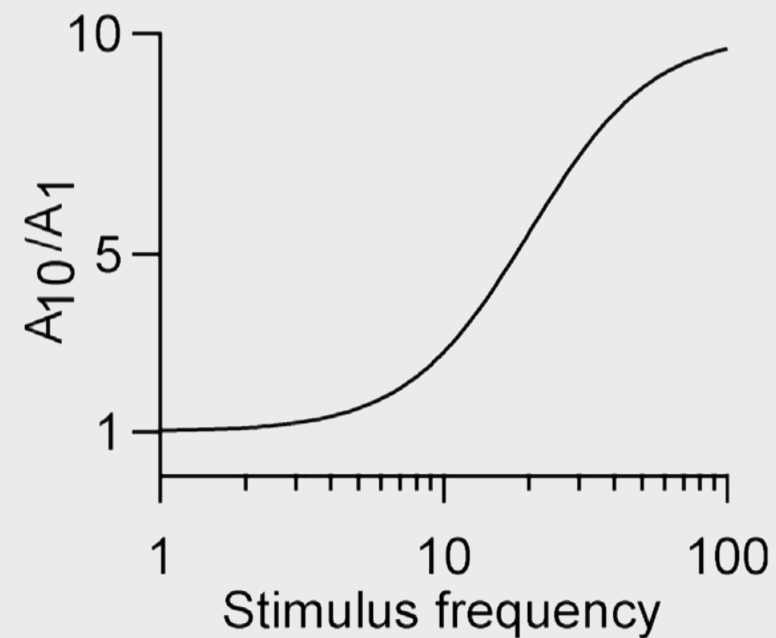
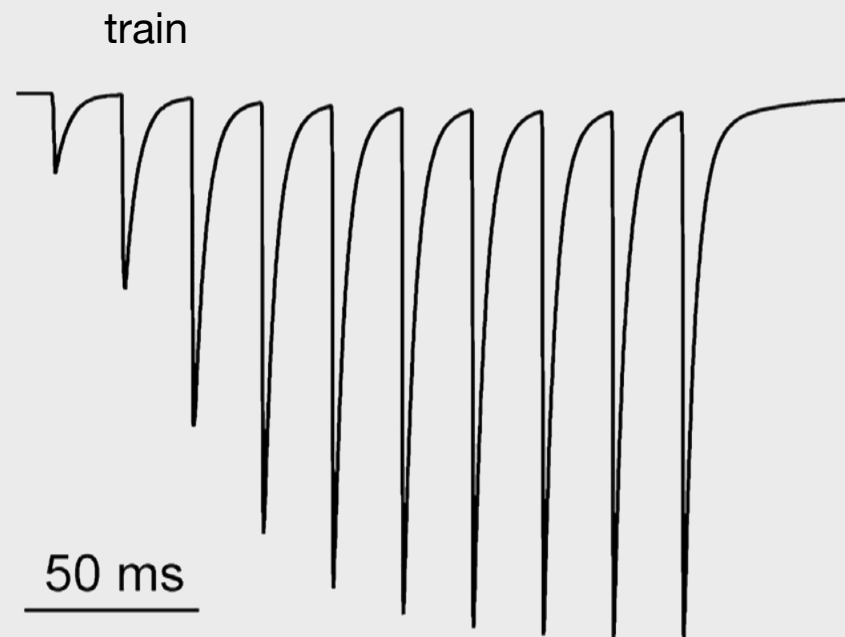
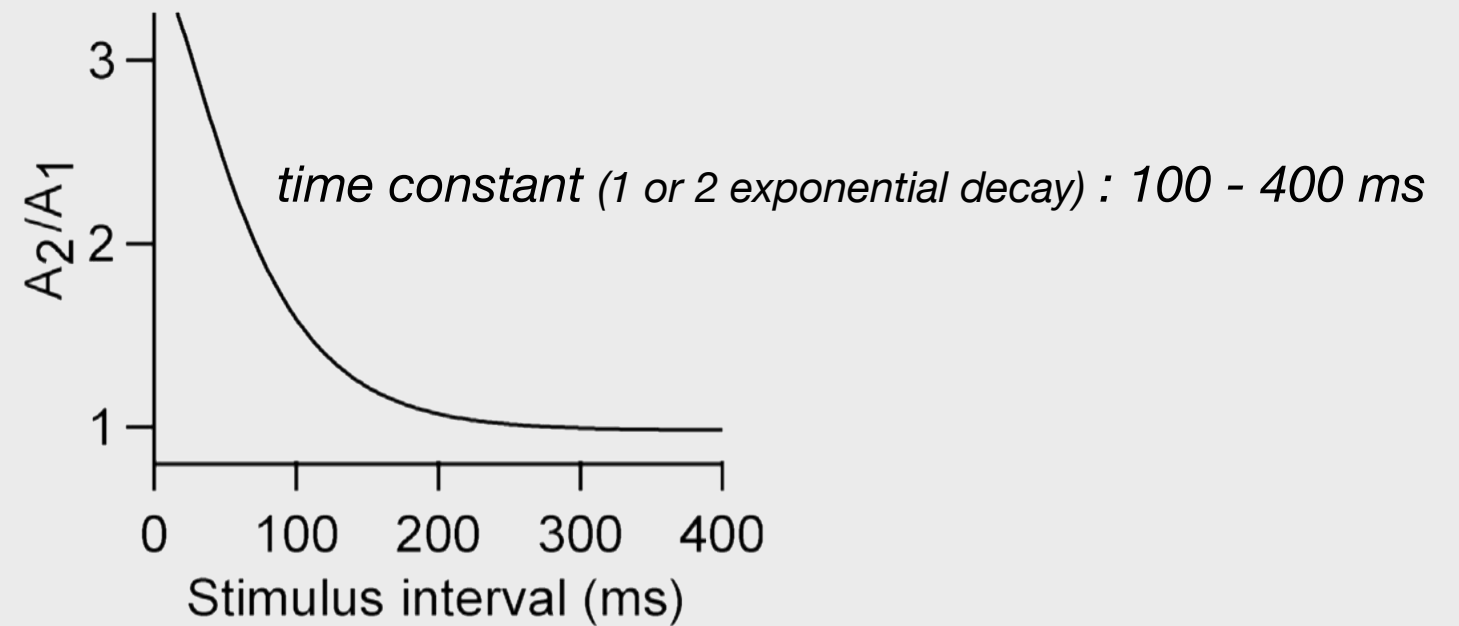
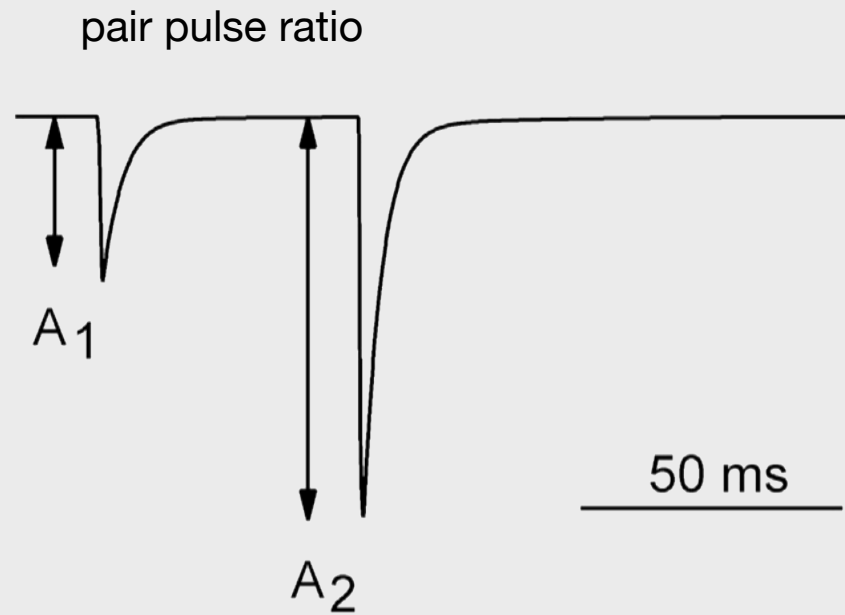


climbing fiber stimulation (75Hz)



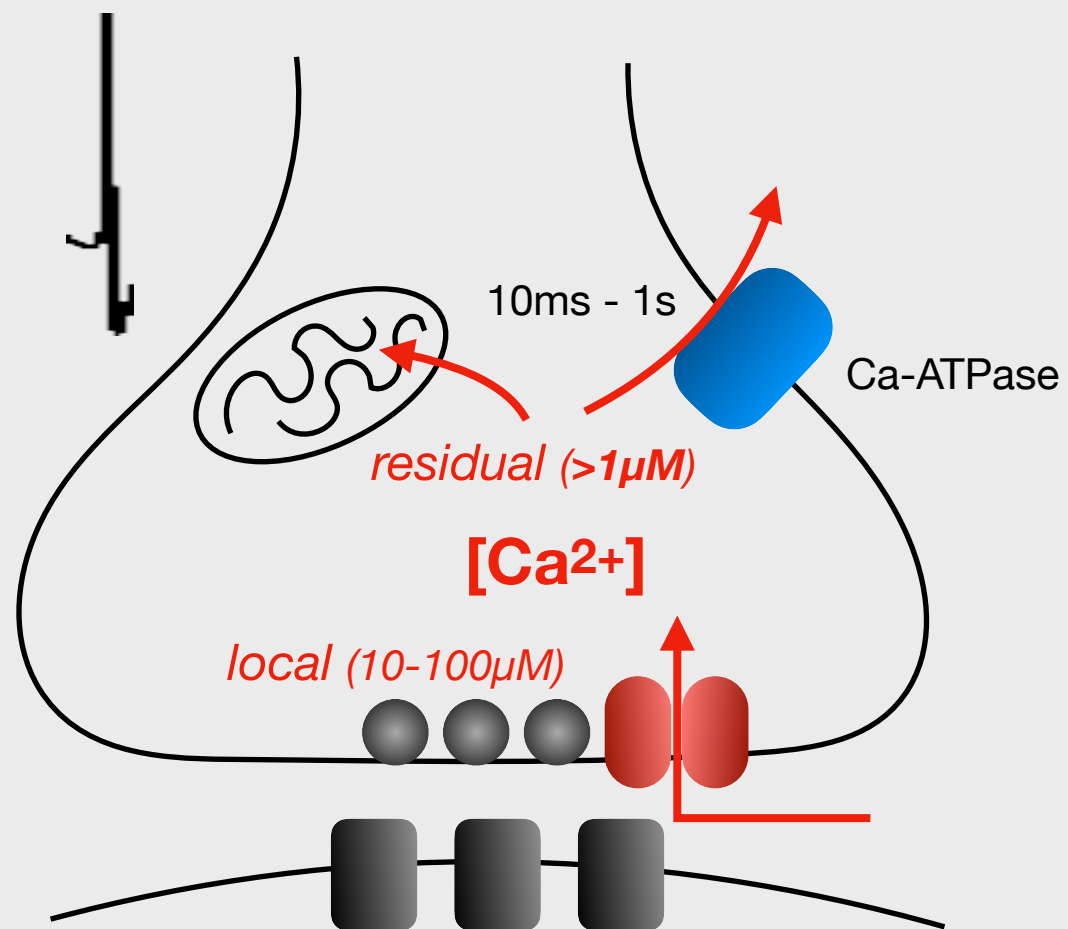
short term plasticity: facilitation

- the second synaptic current can be 5 times larger
- during a high-frequency stimulation train, the amplitude of the synaptic current can be multiplied by 12

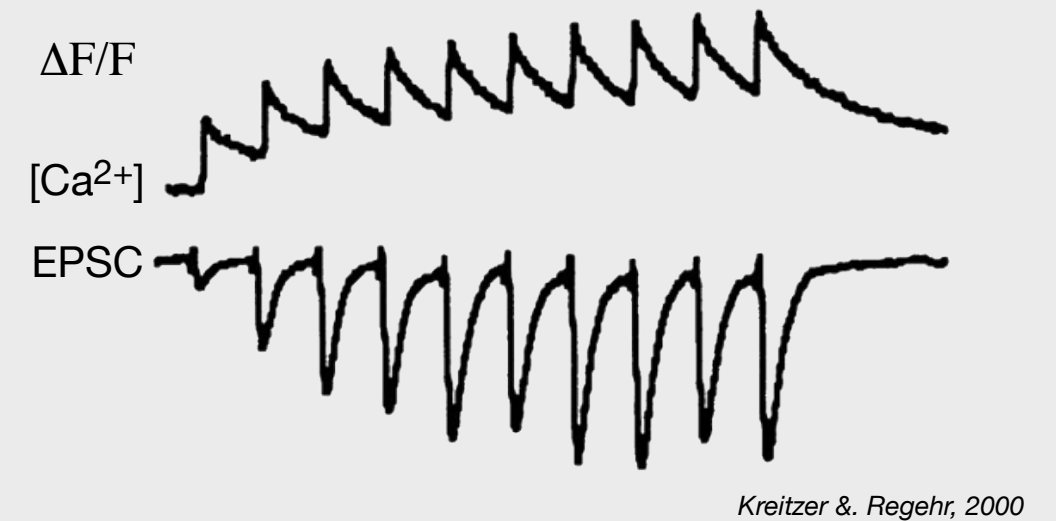


short term plasticity: facilitation

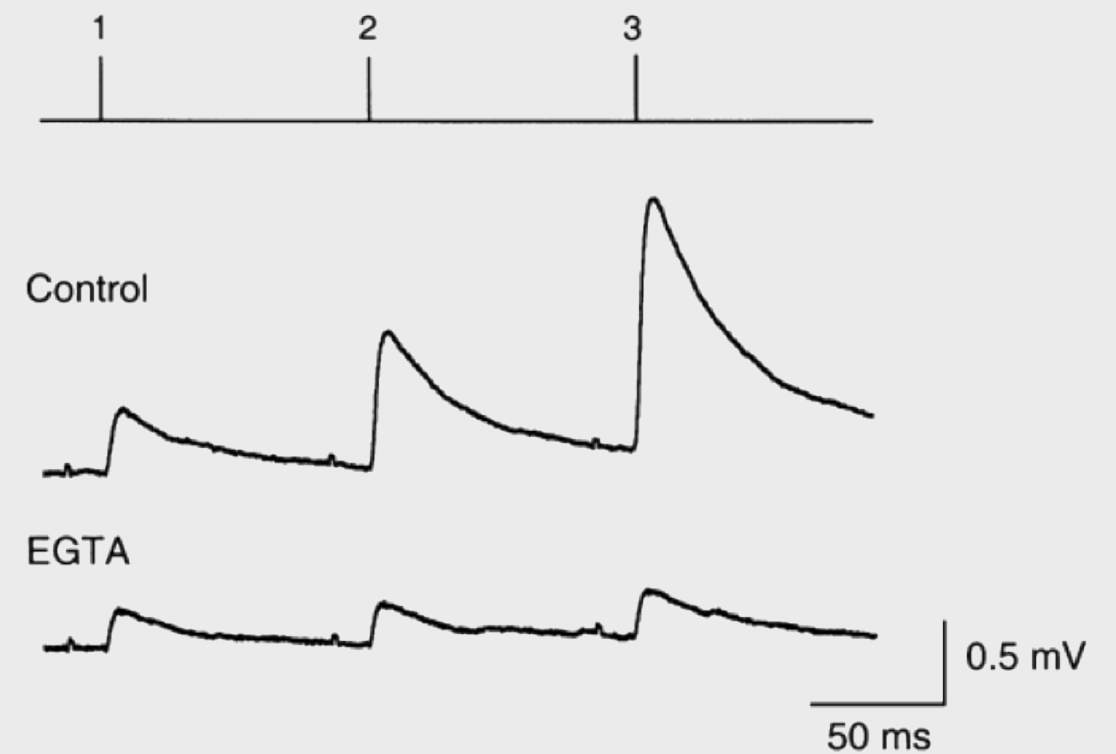
residual calcium



parallel fiber

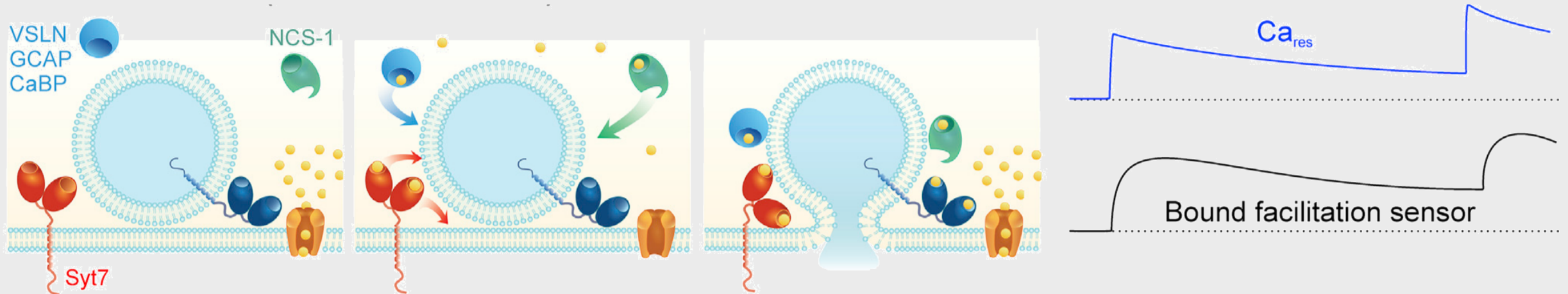


pyramidal cortical neuron -> interneuron



short term plasticity: facilitation

residual calcium



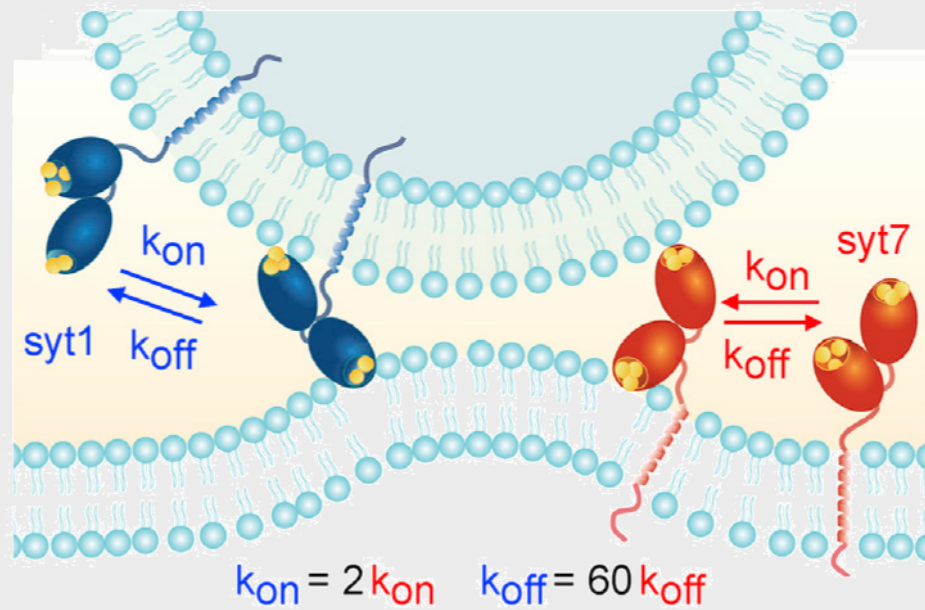
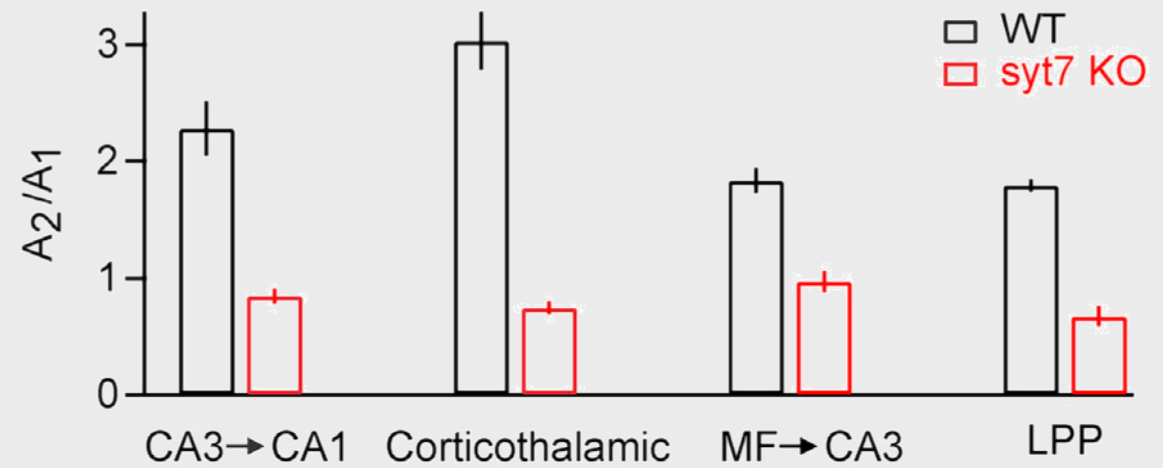
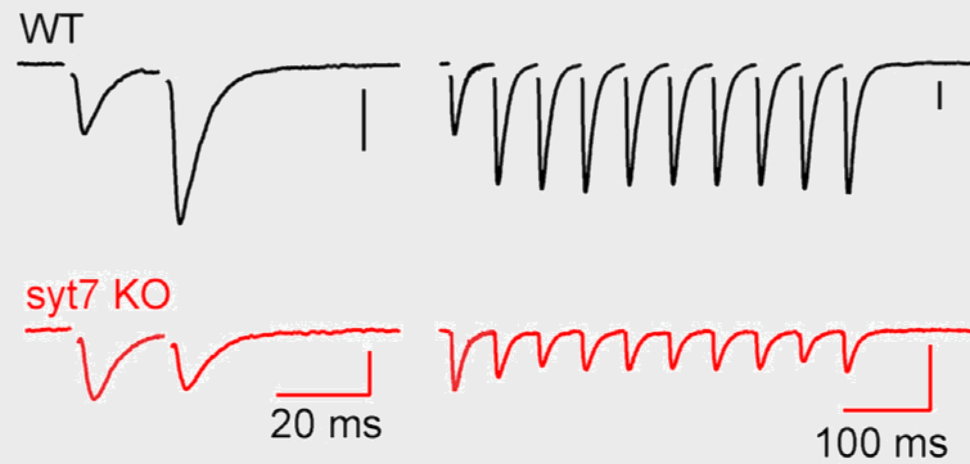
Jackman & Regehr, 2017

Residual Ca^{2+} binds to a facilitation sensor:

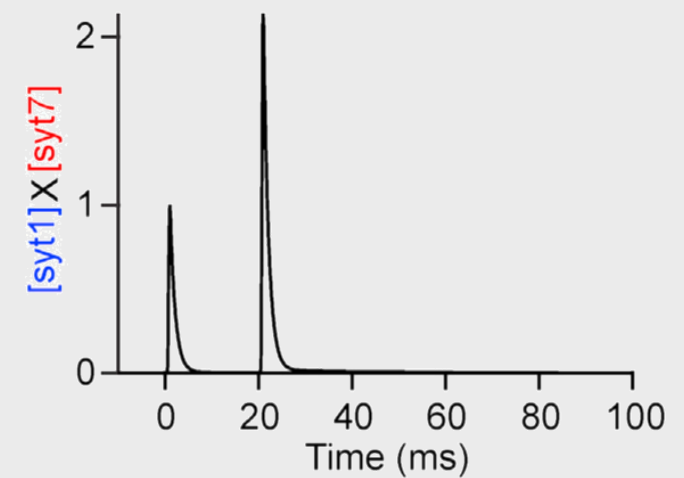
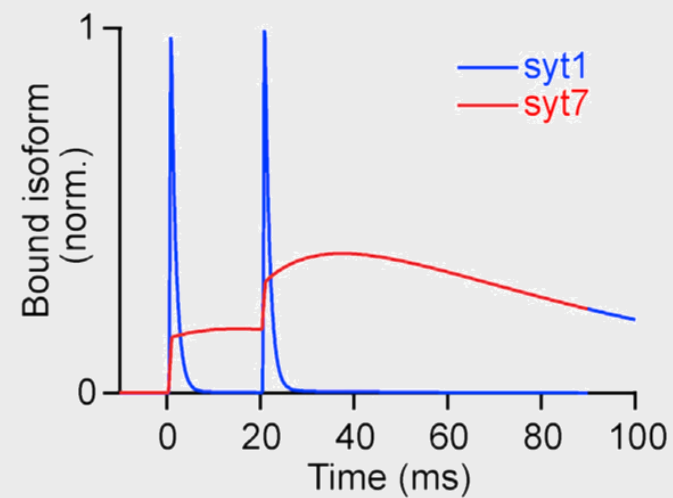
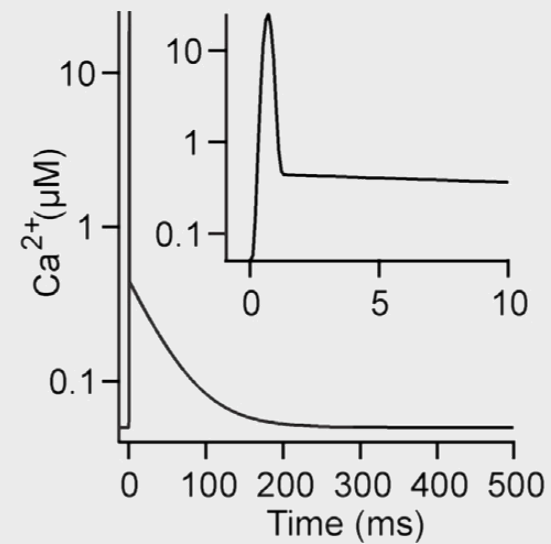
one or multiple high-affinity sensors (Syt7, VSLN, NCS-1...), distinct from Synaptotagmin 1, that bind to residual Ca^{2+} in between APs and raise the release probability by interacting with the fusion machinery

short term plasticity: facilitation

synaptotagmin 7

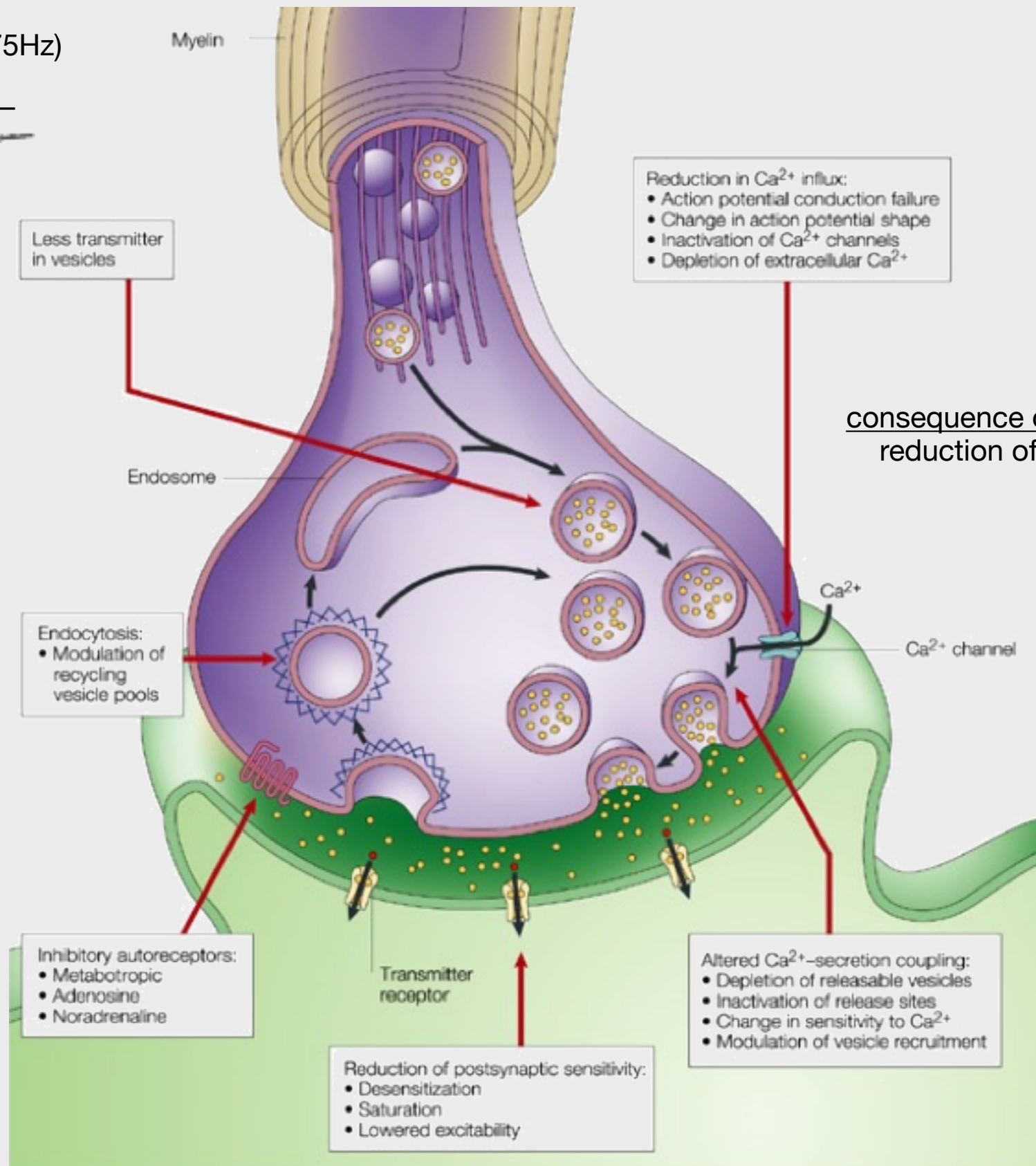
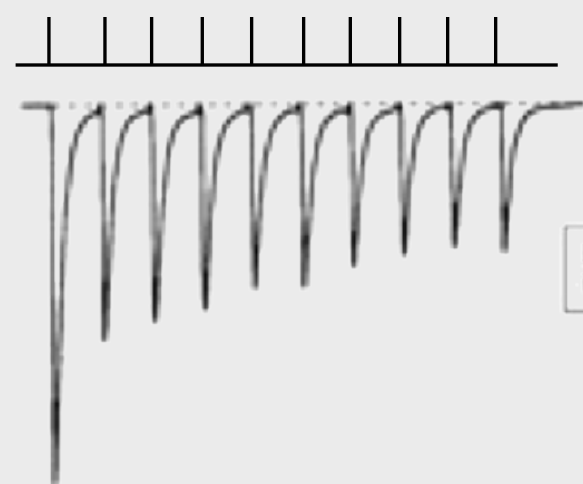


Jackman & Regehr, 2017



short term plasticity: depression

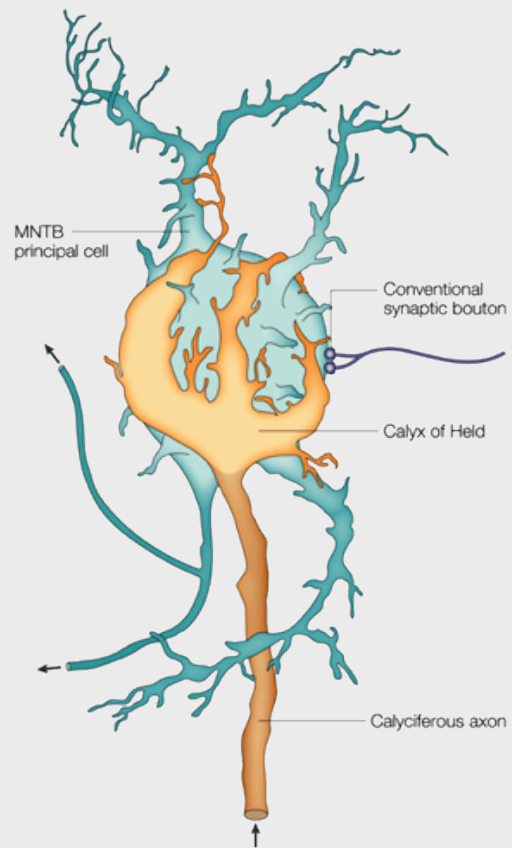
climbing fiber stimulation (75Hz)



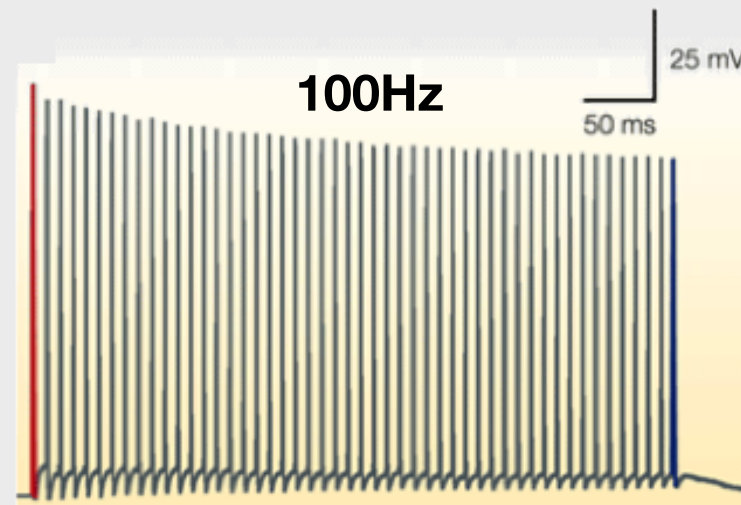
consequence of the pre-synaptic mechanism:
reduction of the neurotransmitter release

short term plasticity: depression

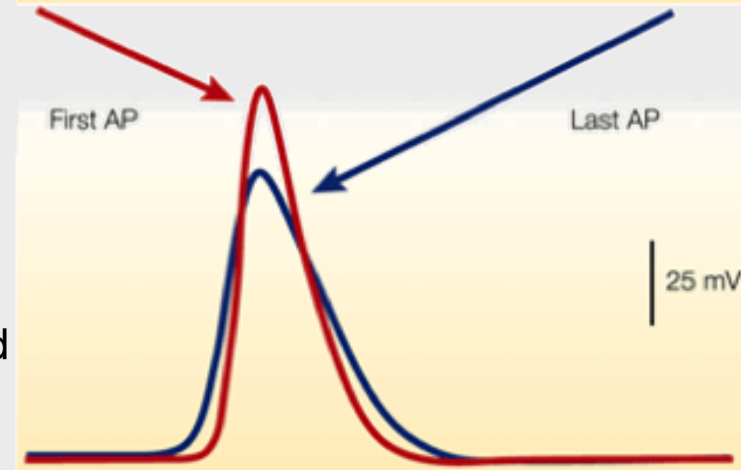
broadening of the APs



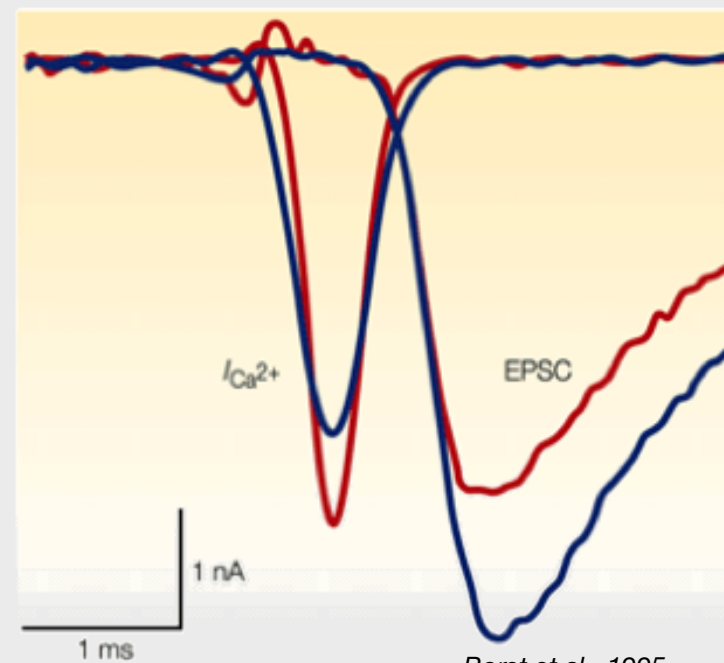
pre-syn. current clamp



pre-syn. voltage command



post-syn. voltage clamp

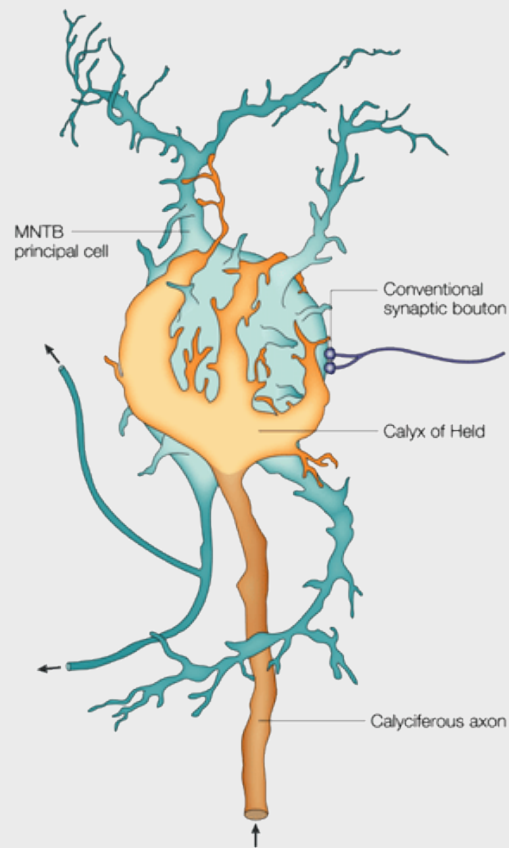


Borst et al., 1995

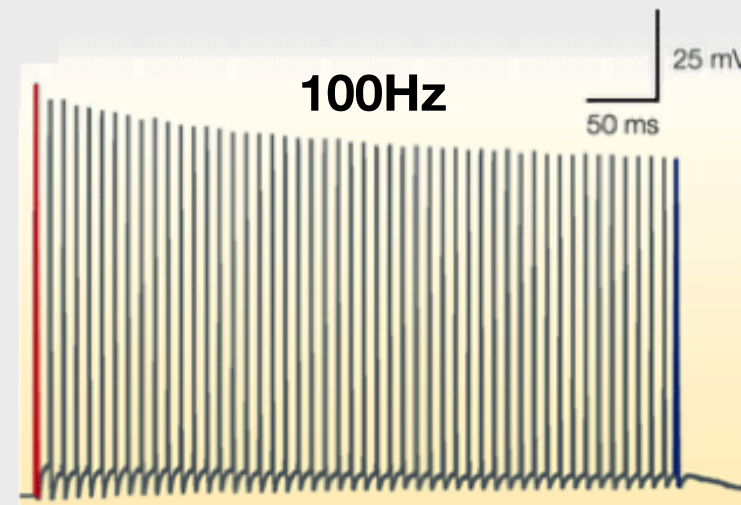
?

short term plasticity: depression

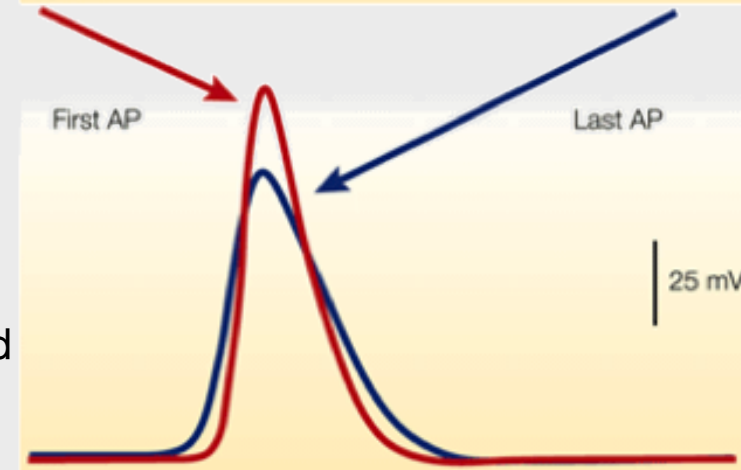
broadening of the APs



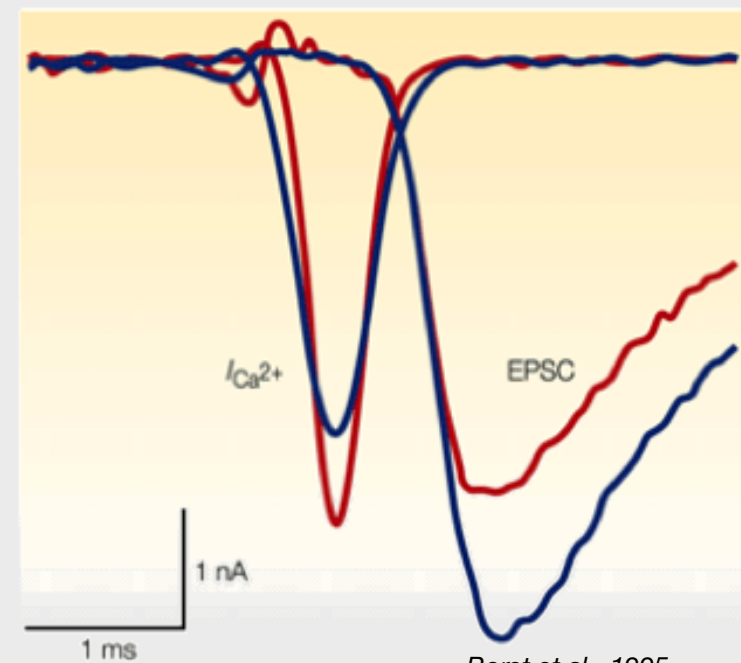
pre-syn. current clamp



pre-syn. voltage command



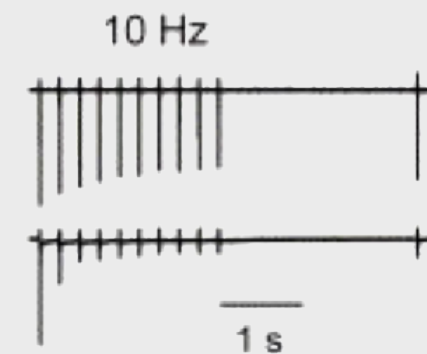
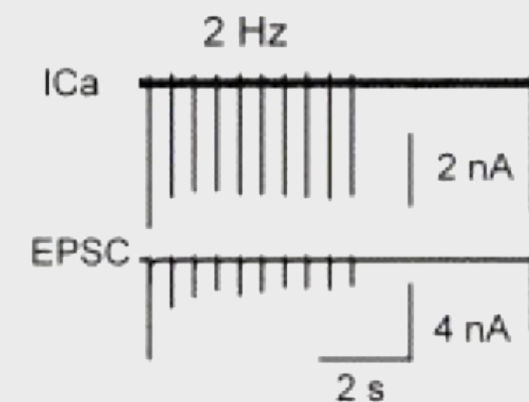
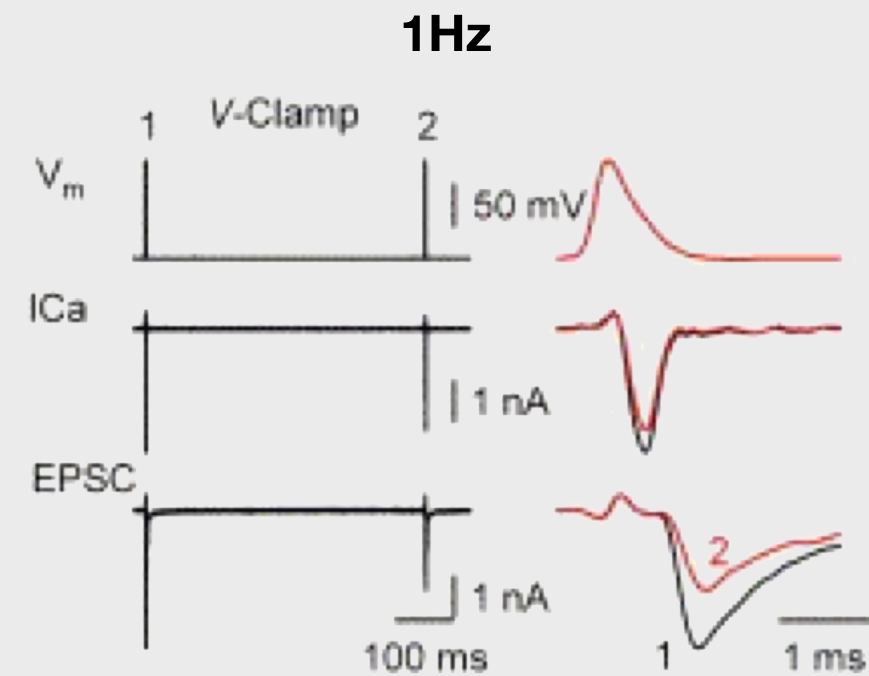
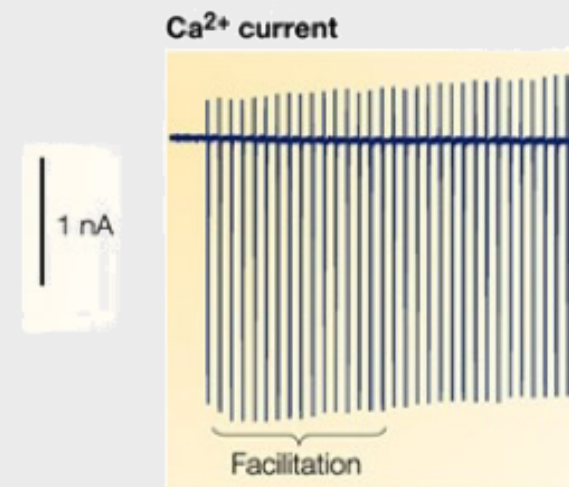
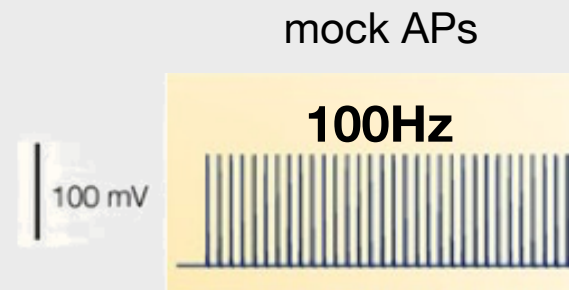
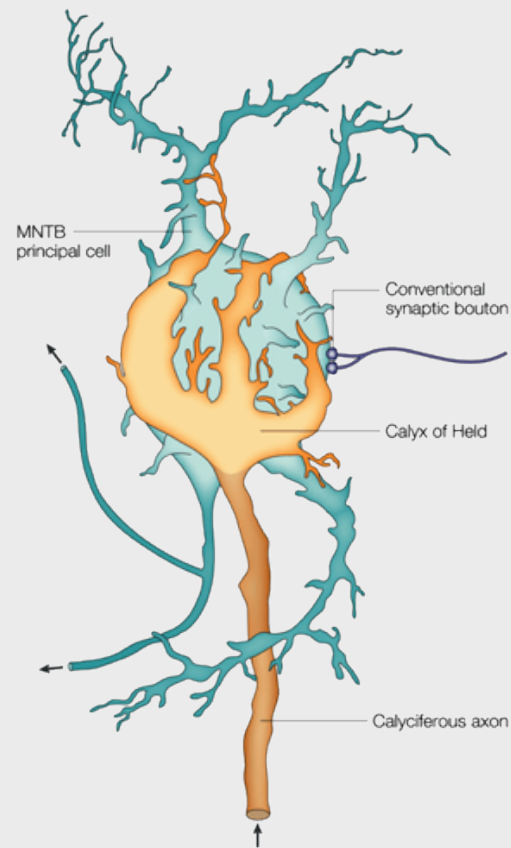
post-syn. voltage clamp



- AP and Ca^{2+} current amplitude are smaller
- total Ca^{2+} entry is larger

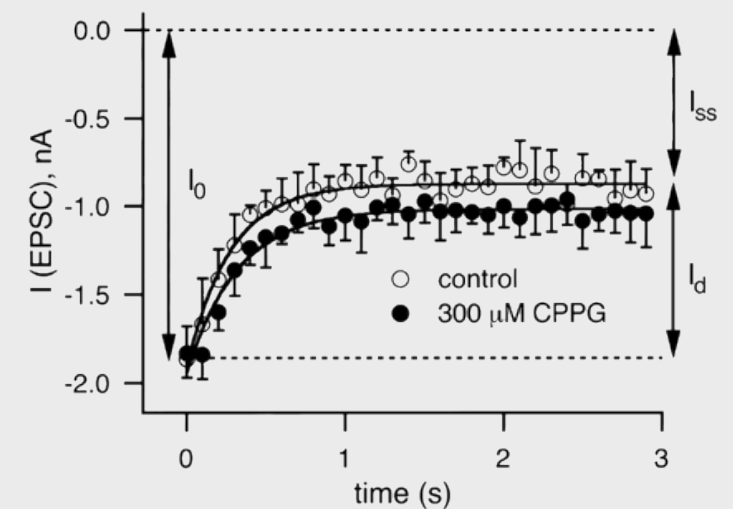
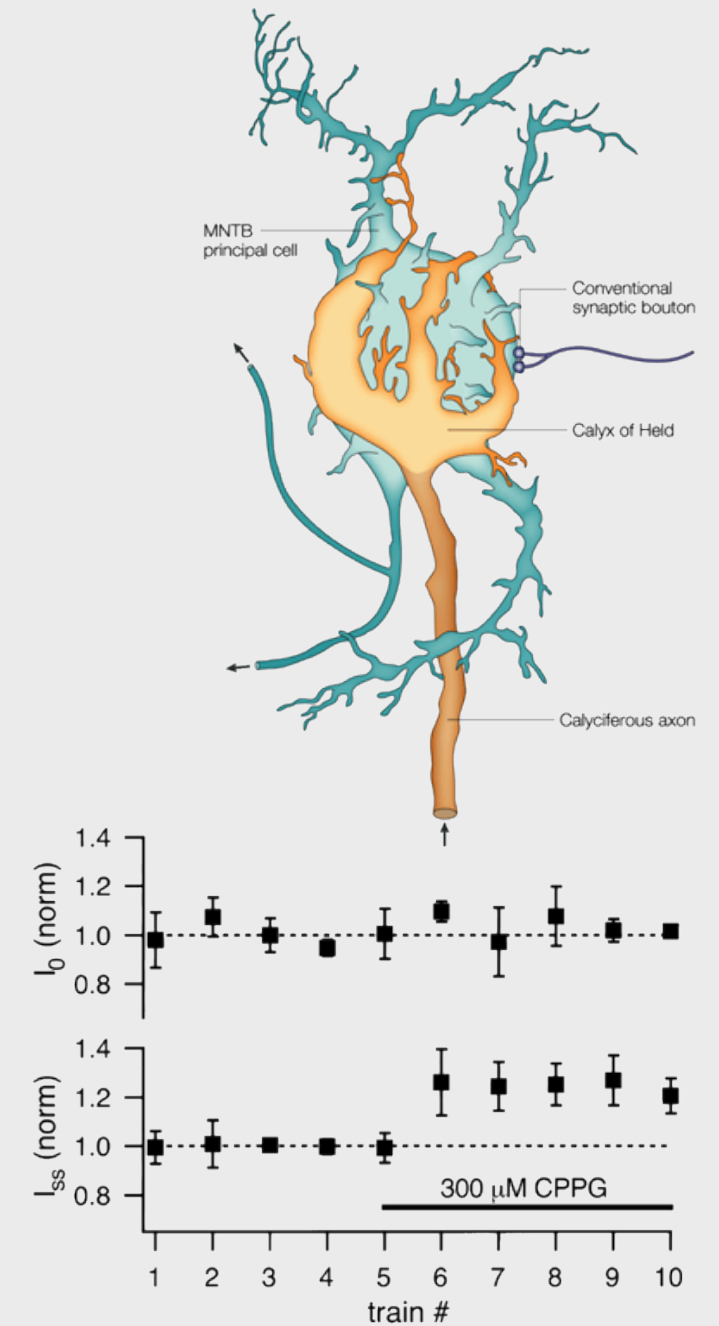
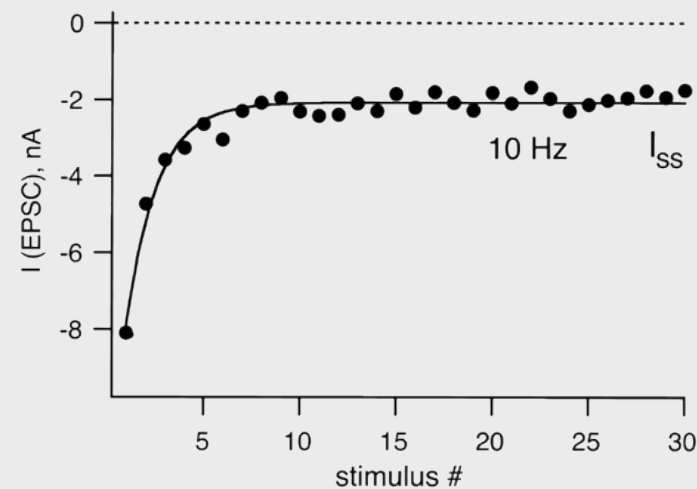
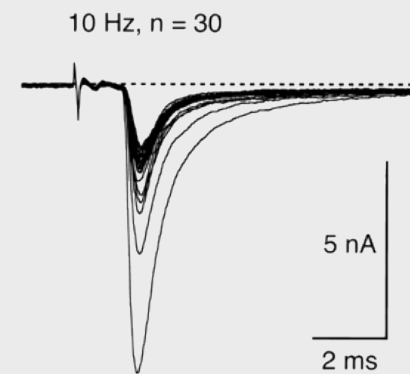
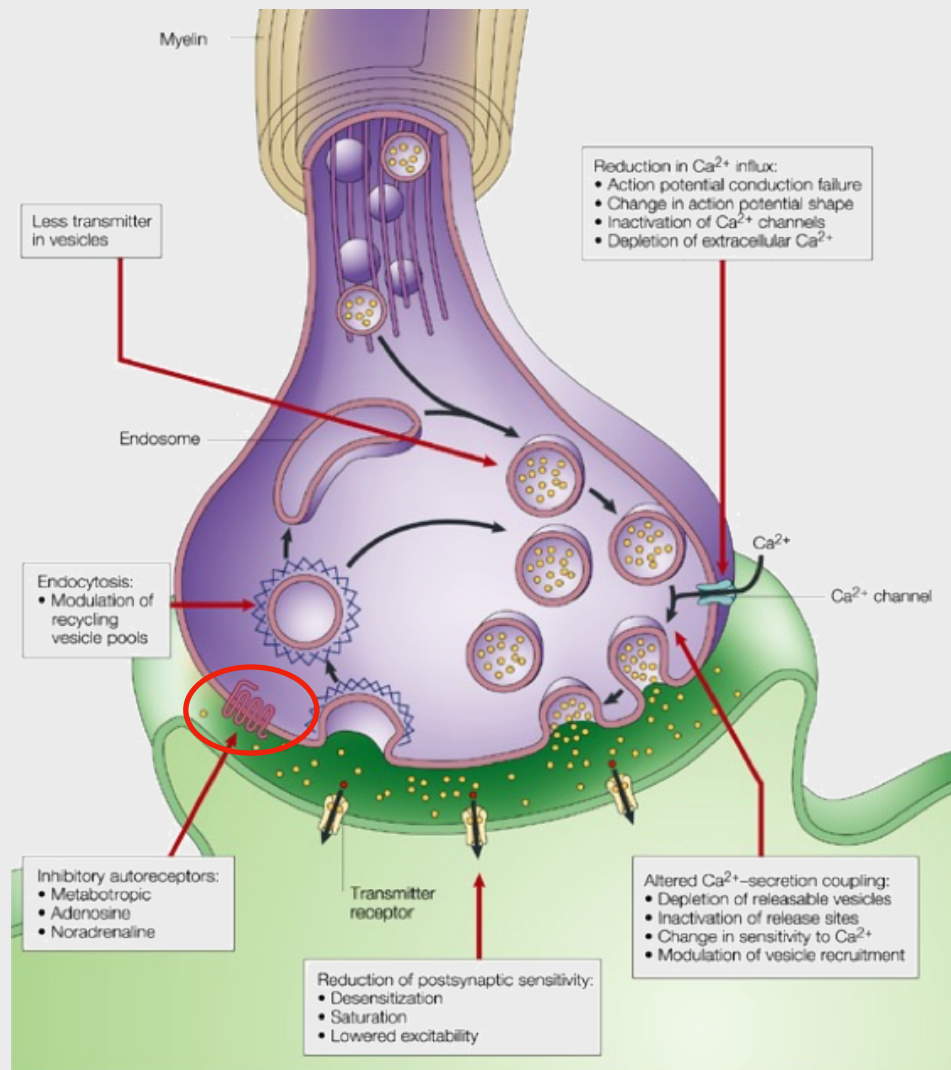
short term plasticity: depression

inactivation of calcium channels (could depend on the AP frequency)



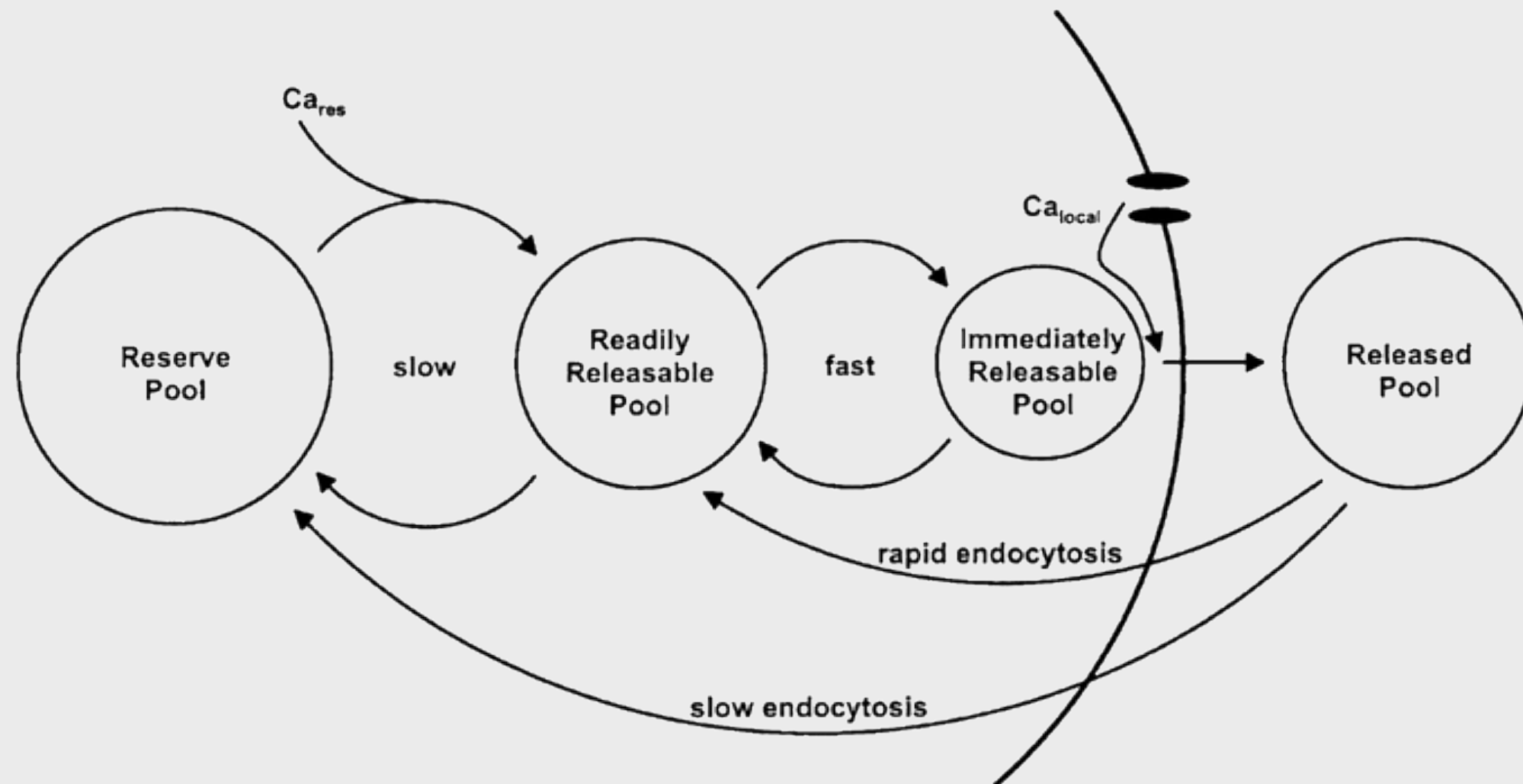
short term plasticity: depression

pre-synaptic metabotropic autoreceptors



short term plasticity: depression

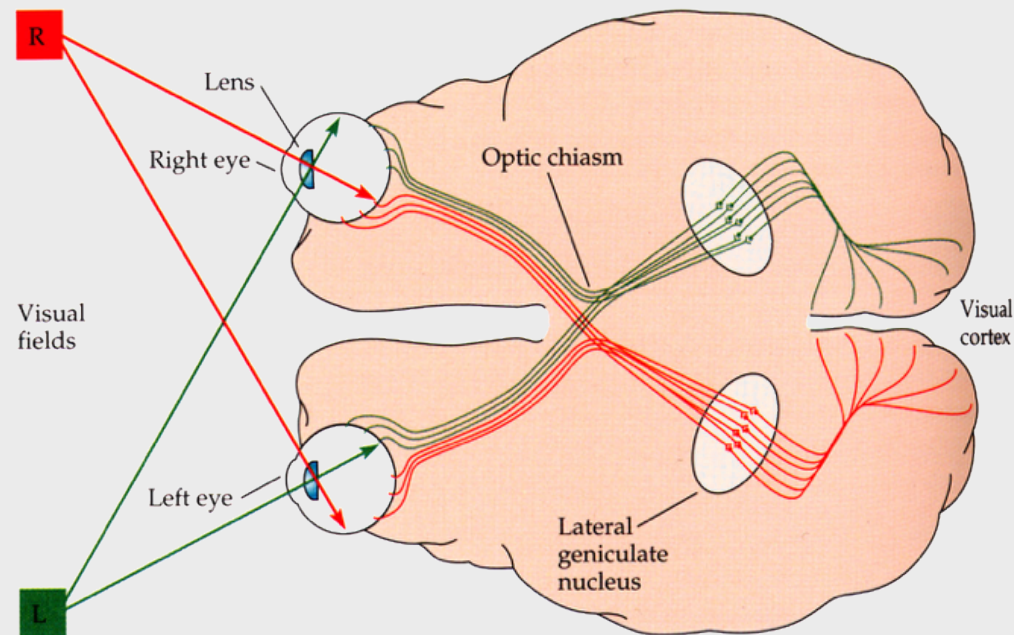
depletion of the Readily Releasable Pool



pre-synaptic mechanism common to all synapses

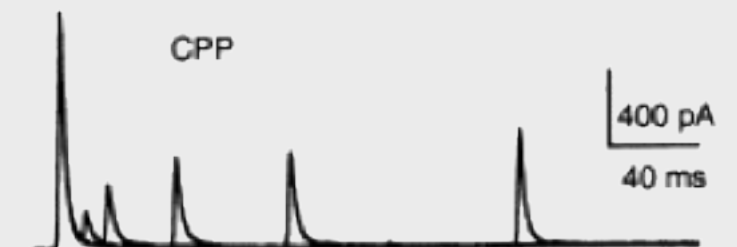
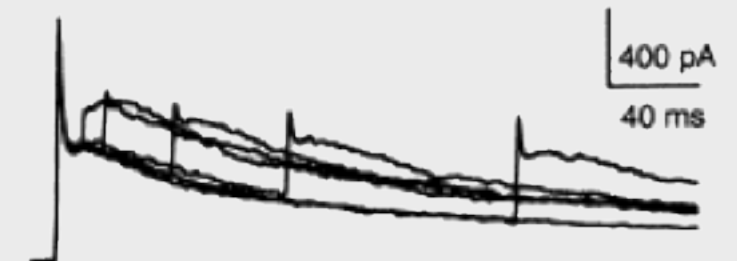
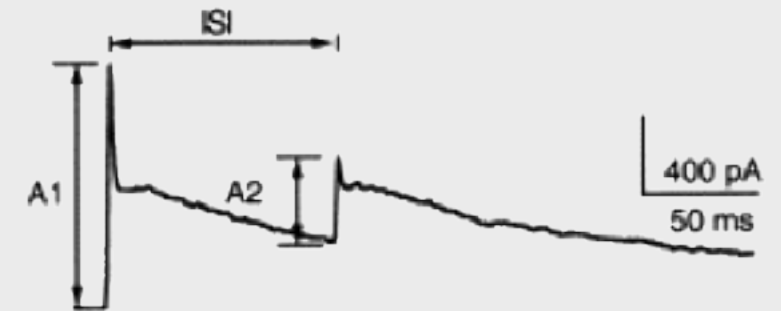
short term plasticity: depression

post-synaptic mechanisms

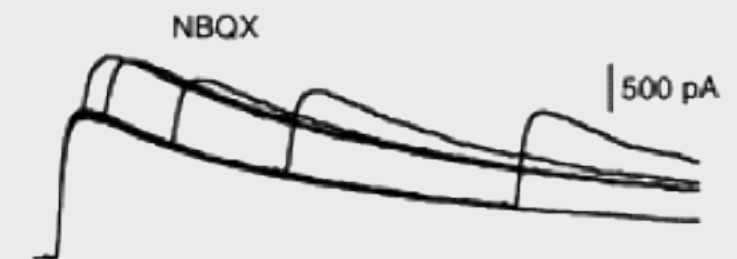


V holding: +40mV

EPSC AMPA

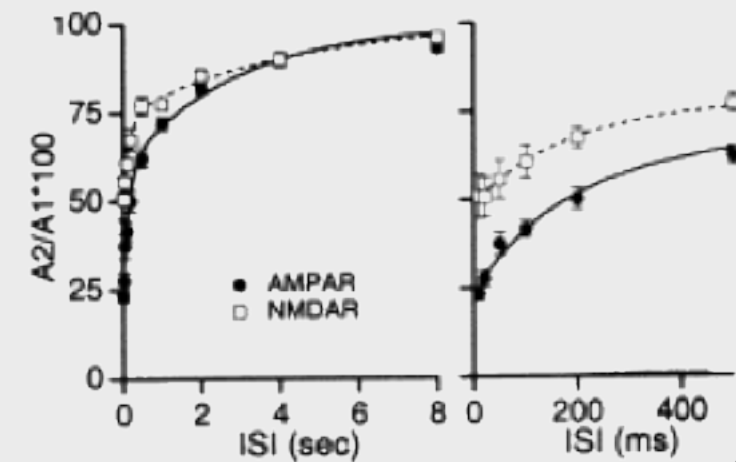


EPSC NMDA



the pair pulse depression is different for AMPA and NMDA currents when considering short ISIs

?

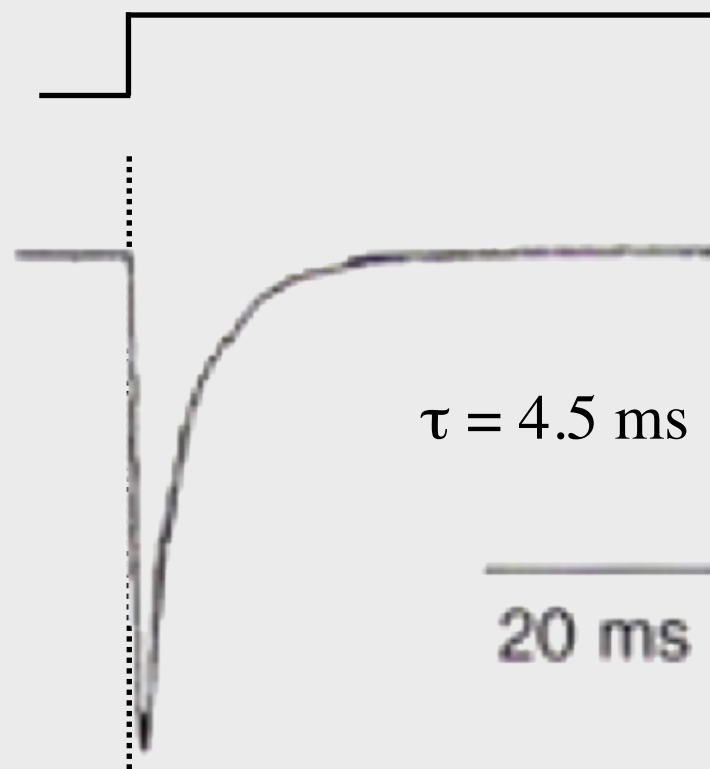


short term plasticity: depression

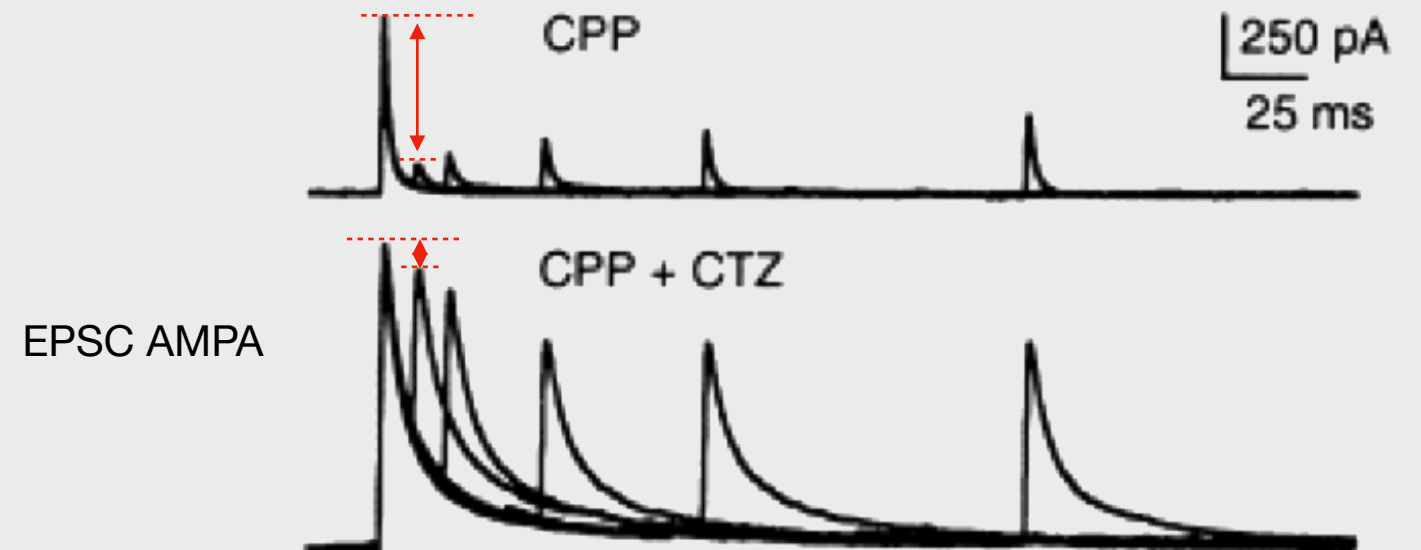
post-synaptic mechanisms

desensitization

sustained application of glutamate



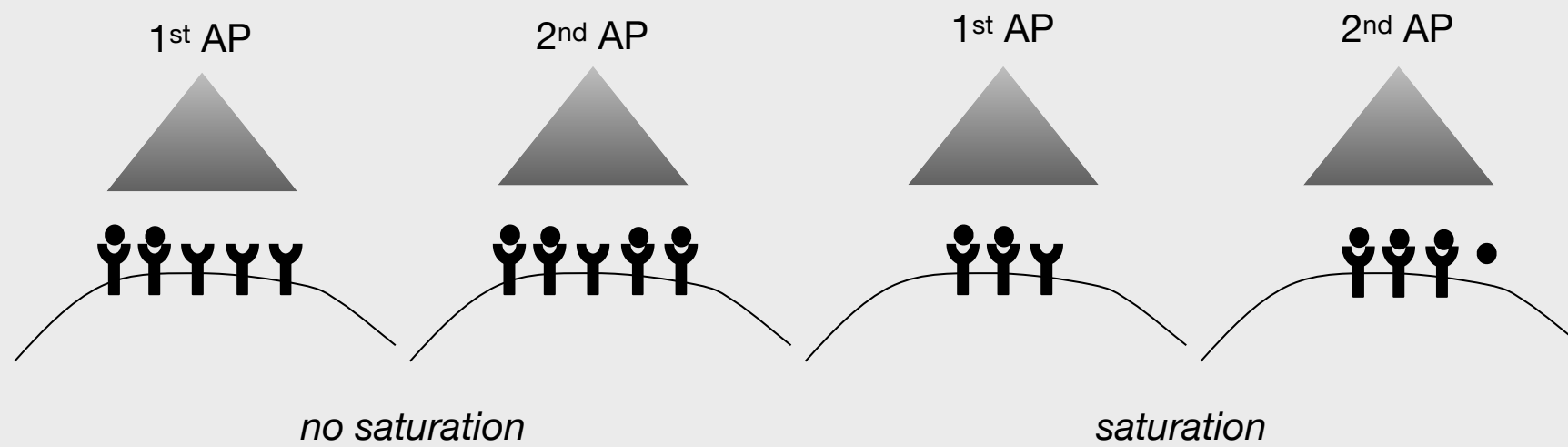
V holding: +40mV



Cyclothiazide (CTZ) reduces AMPA-R desensitization

short term plasticity: depression

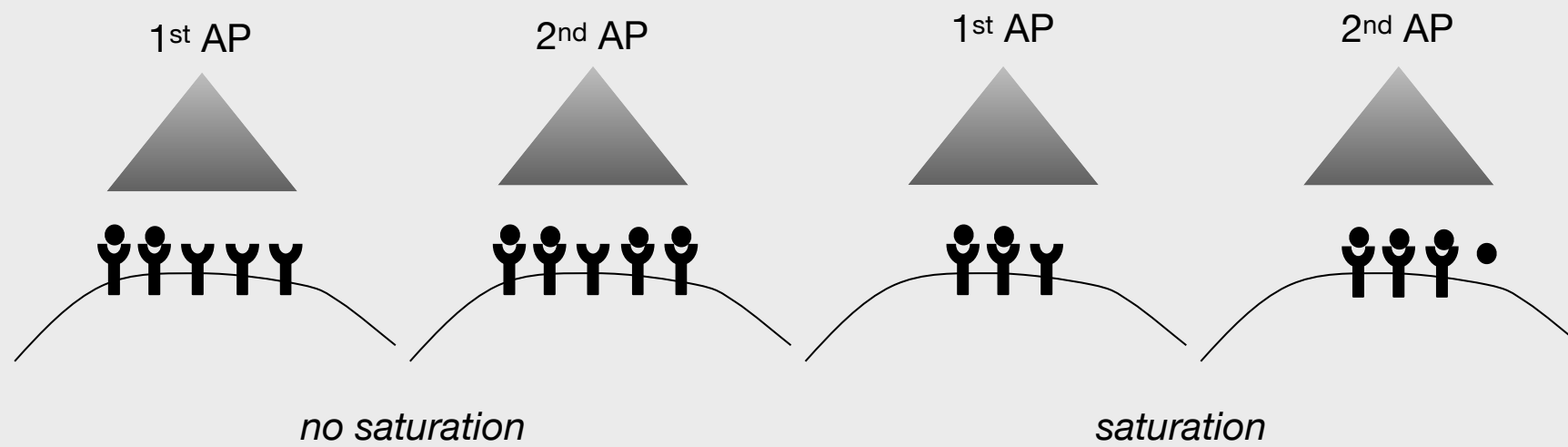
post-synaptic mechanisms



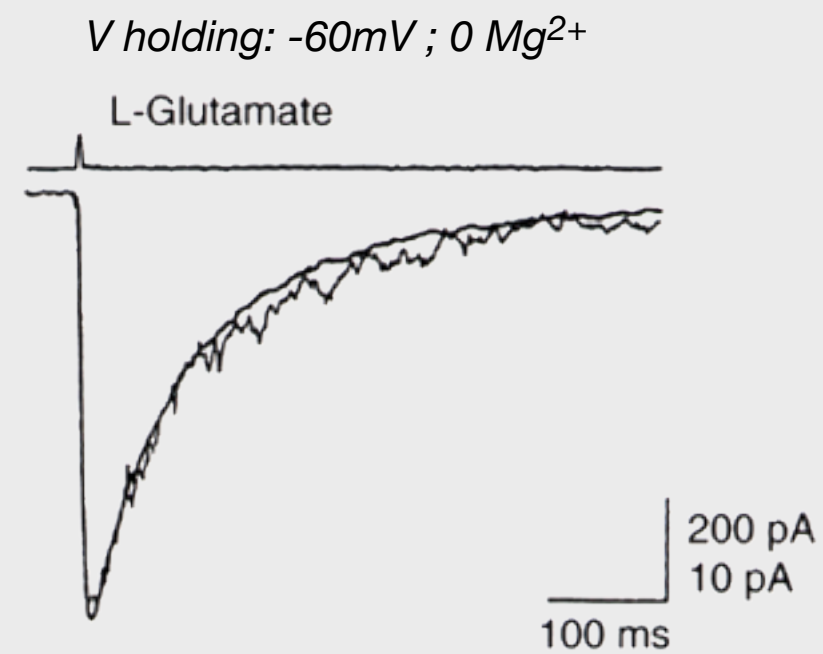
Saturation is only possible if the receptor deactivation is slow.

short term plasticity: depression

post-synaptic mechanisms

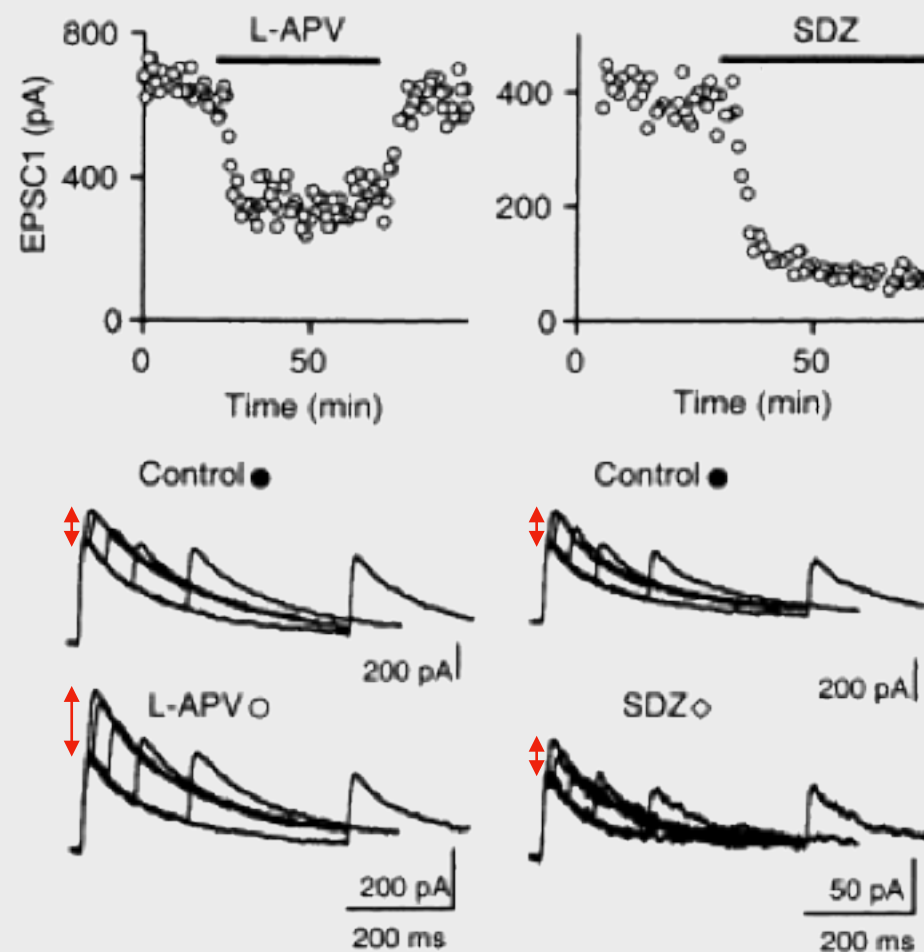
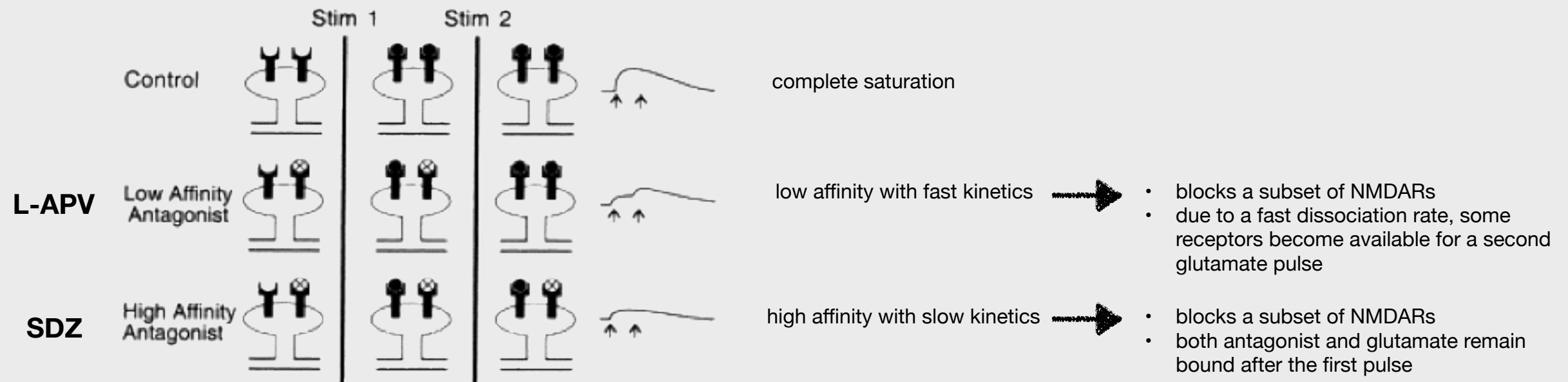


NMDA-R



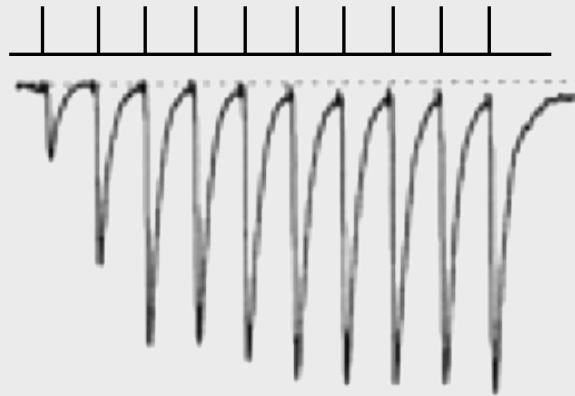
short term plasticity: depression

post-synaptic mechanisms



short term plasticity: facilitation and depression

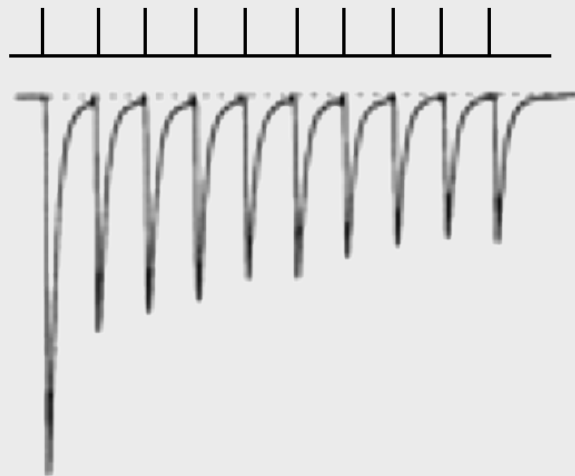
parallel fiber stimulation (75Hz)



low probability of release :

- residual calcium

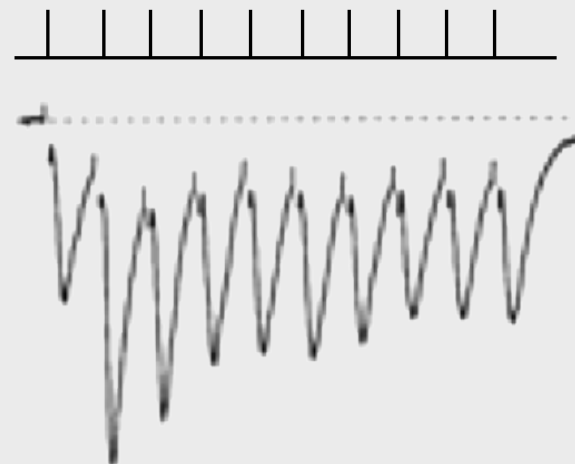
climbing fiber stimulation (75Hz)



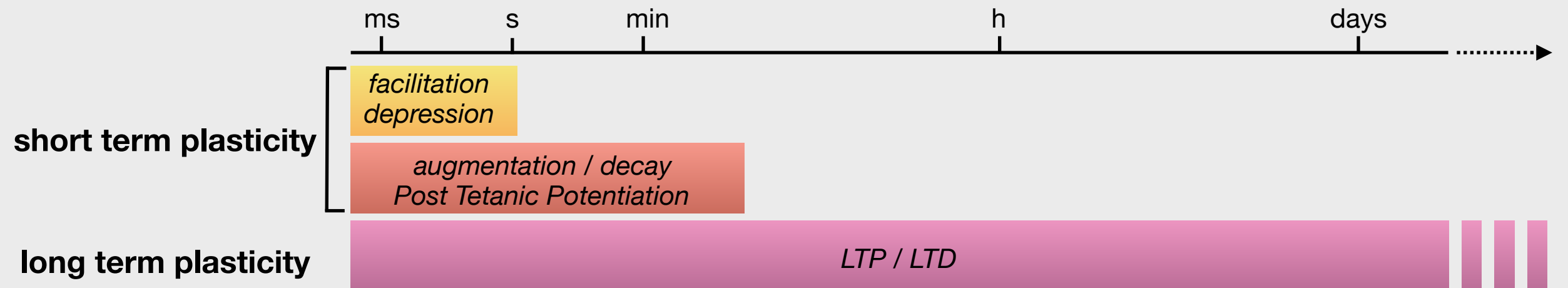
high probability of release :

- depletion of the RRP
- change in AP shape
- Ca^{2+} channel inactivation
- pre-synaptic autoreceptors
- post-synaptic desensitization/saturation

Schaffer collateral stimulation (75Hz)

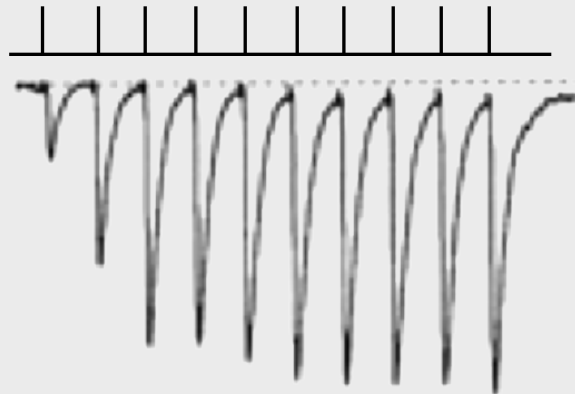


different time scales of synaptic plasticity

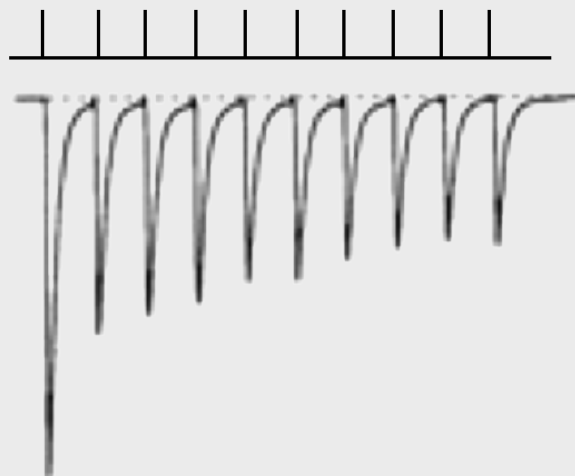


different time scales of synaptic plasticity

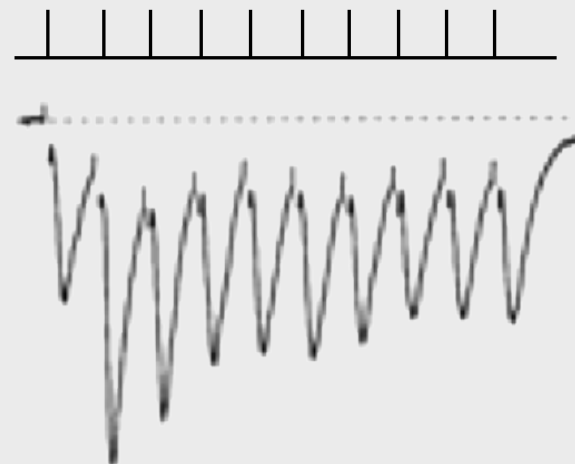
parallel fiber stimulation (75Hz)



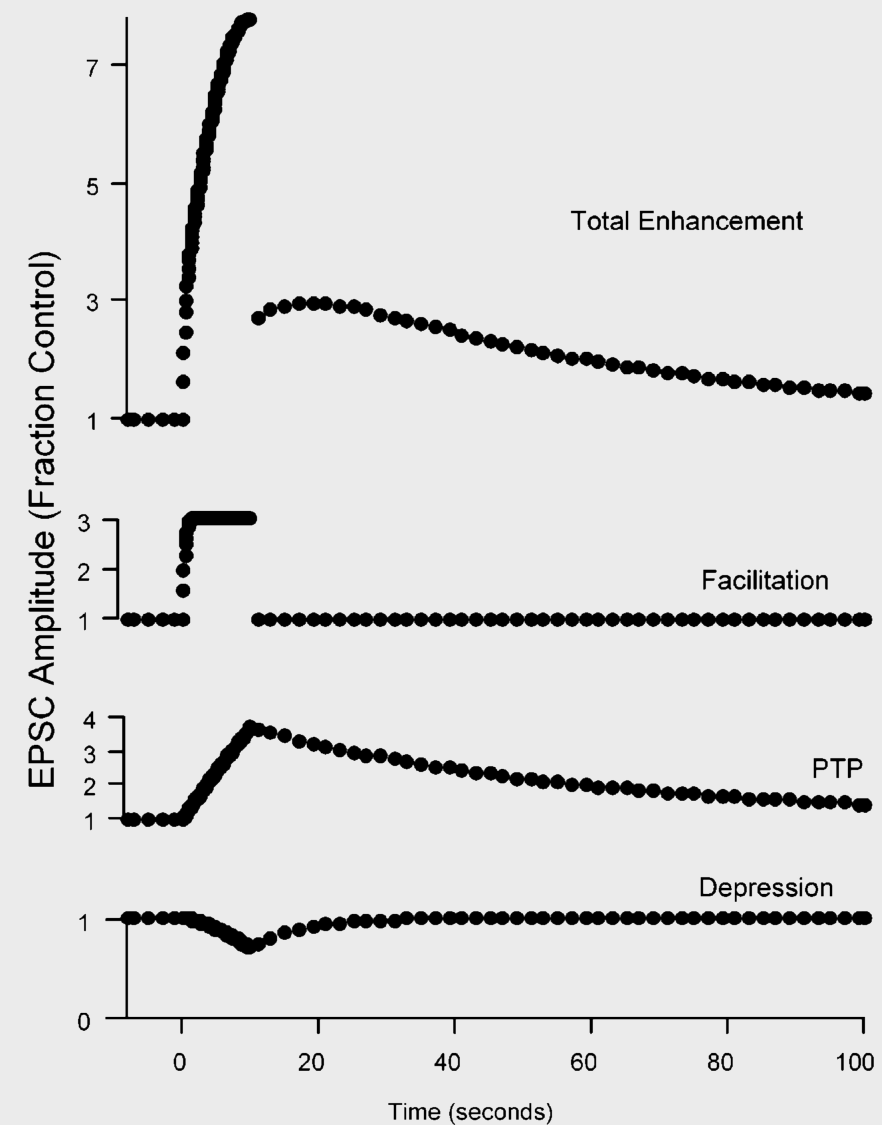
climbing fiber stimulation (75Hz)



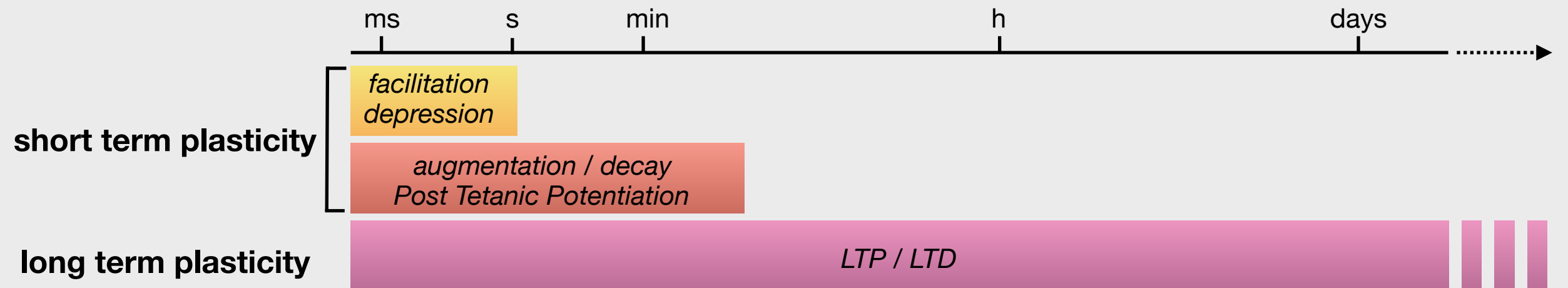
Schaffer collateral stimulation (75Hz)



prolonged stimulation (a few seconds) can lead to a sustained increase in the probability of release lasting a few minutes (post-tetanic potentiation).

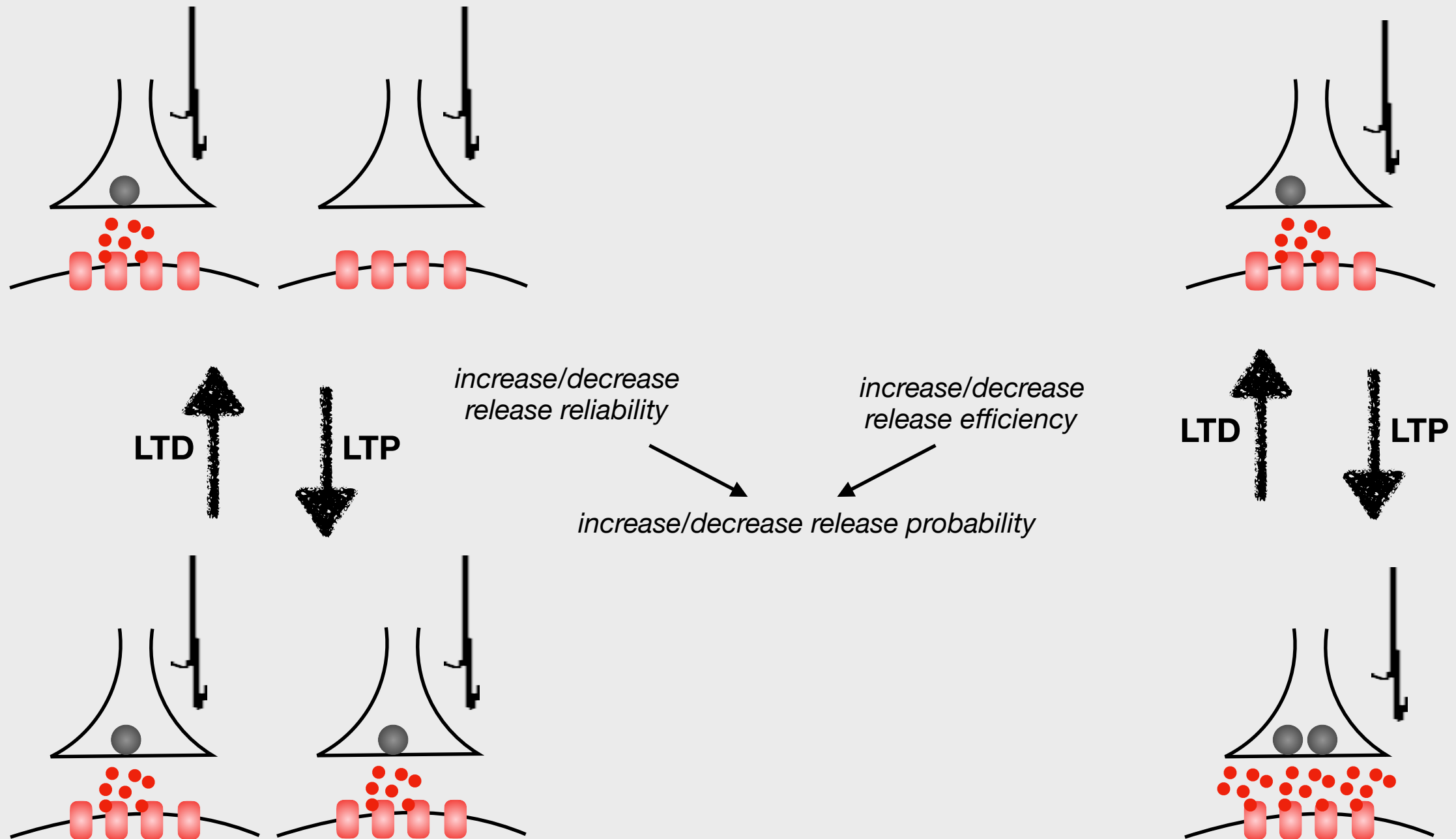


different time scales of synaptic plasticity



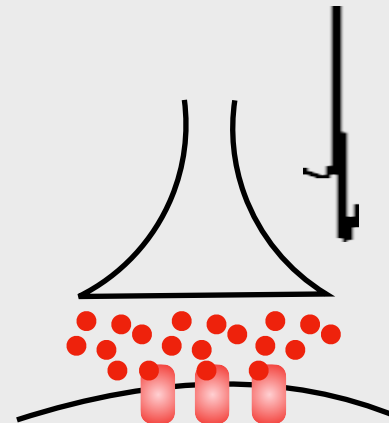
long term plasticity

pre-synaptic mechanism

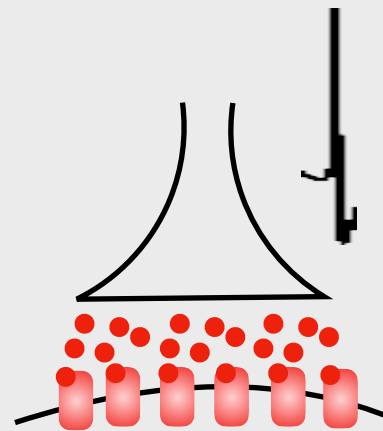
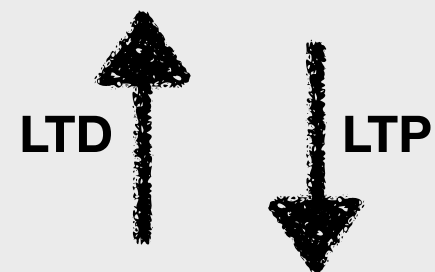


long term plasticity

post-synaptic mechanism

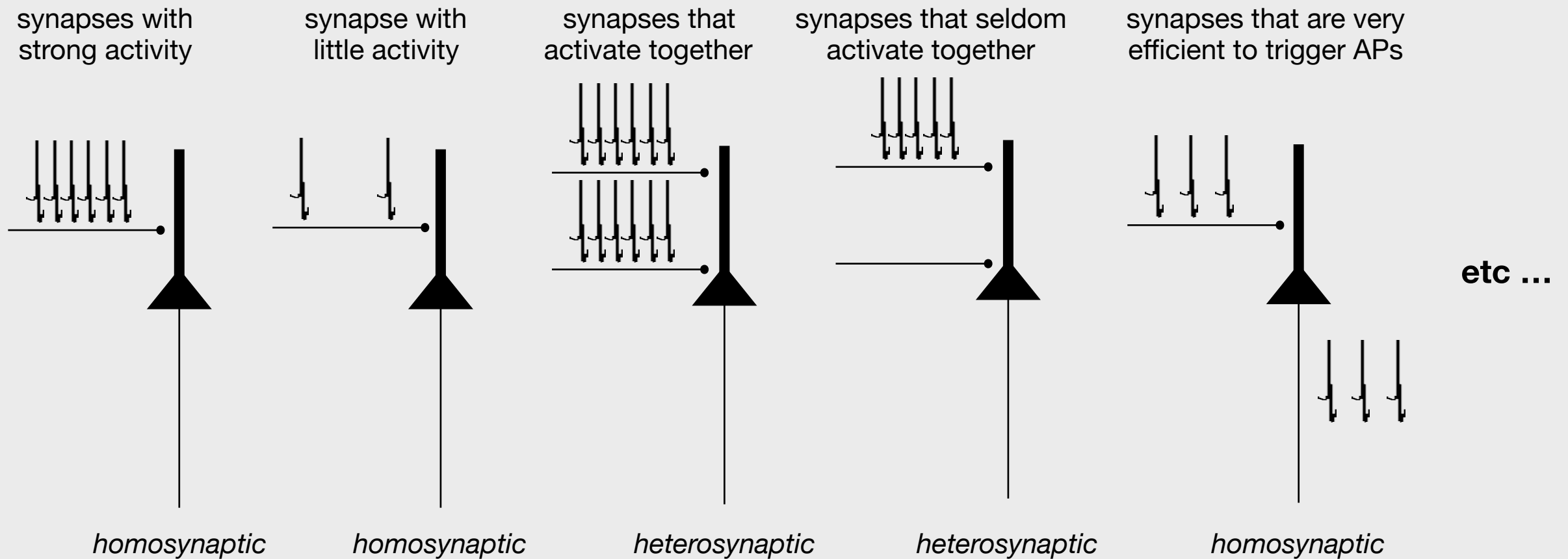


*increase/decrease the number and/or the efficacy
of the receptors*



long term plasticity

which synapse should be modified ?

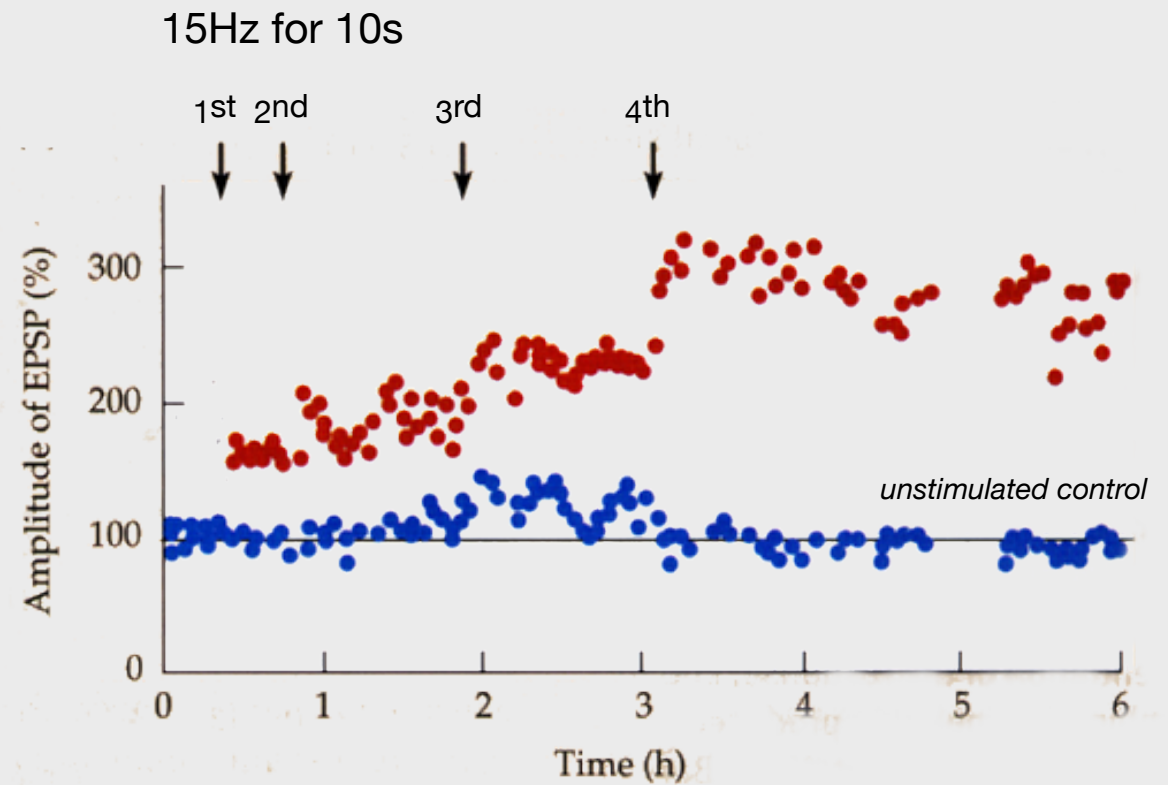
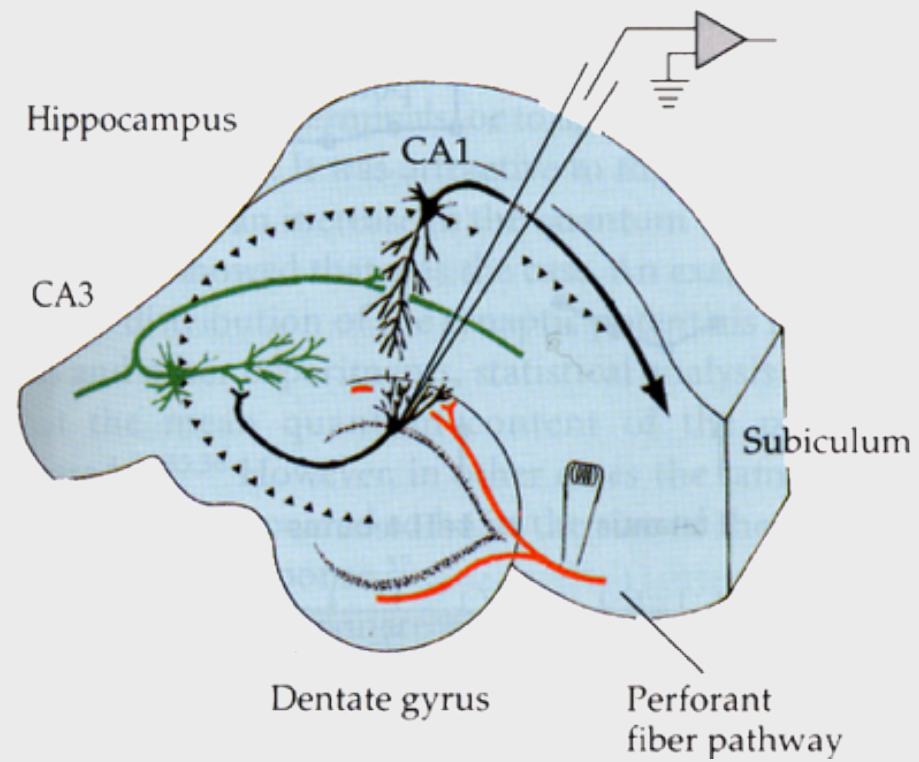


triggering long terme plasticity requires :

- *an intracellular messenger that links changes in Vm to intracellular pathways*
- *in many cases, coincidence detection*
- *in some cases, a retrograde messenger produced in the post-synaptic neuron inducing pre-synaptic plasticity*

long term plasticity

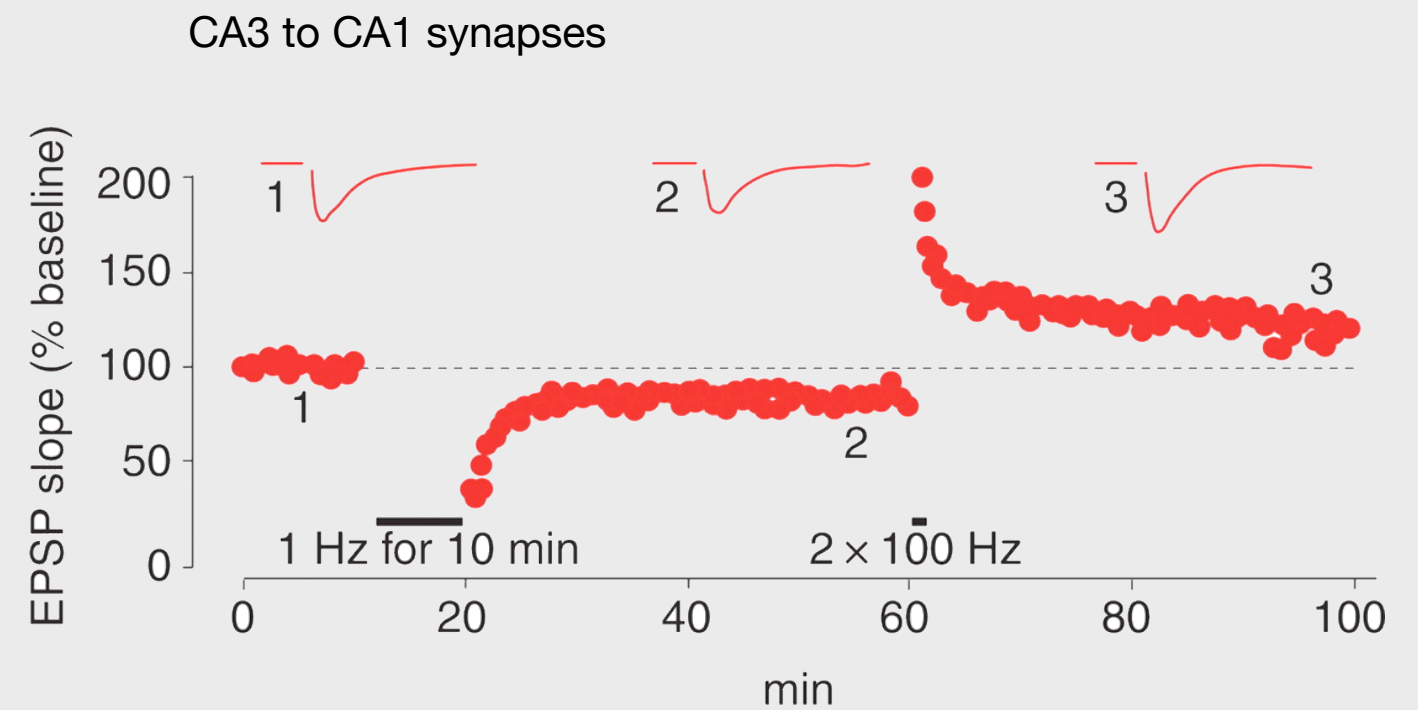
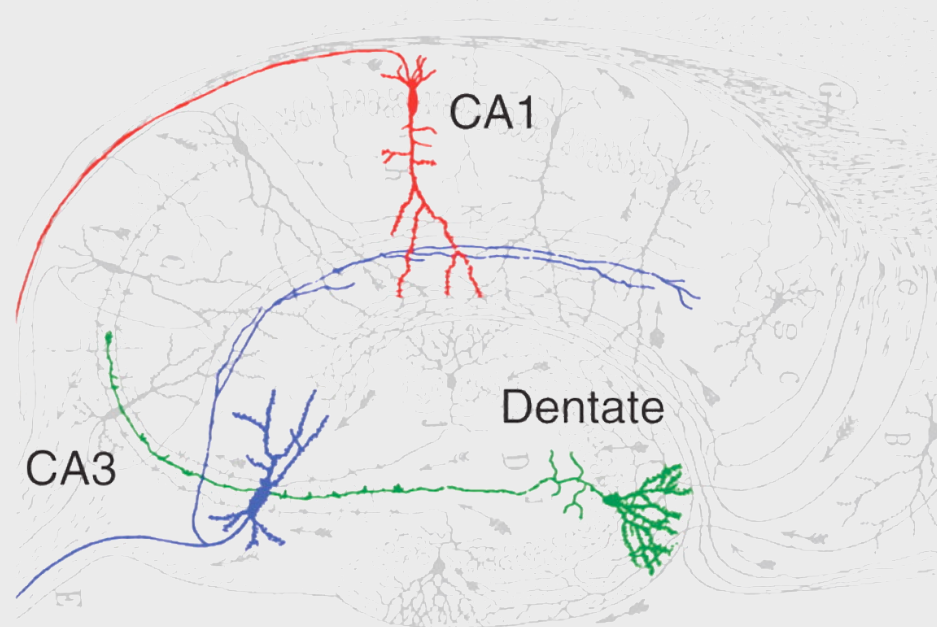
exemple of homosynaptic plasticity



Bliss & Lomo, 1973

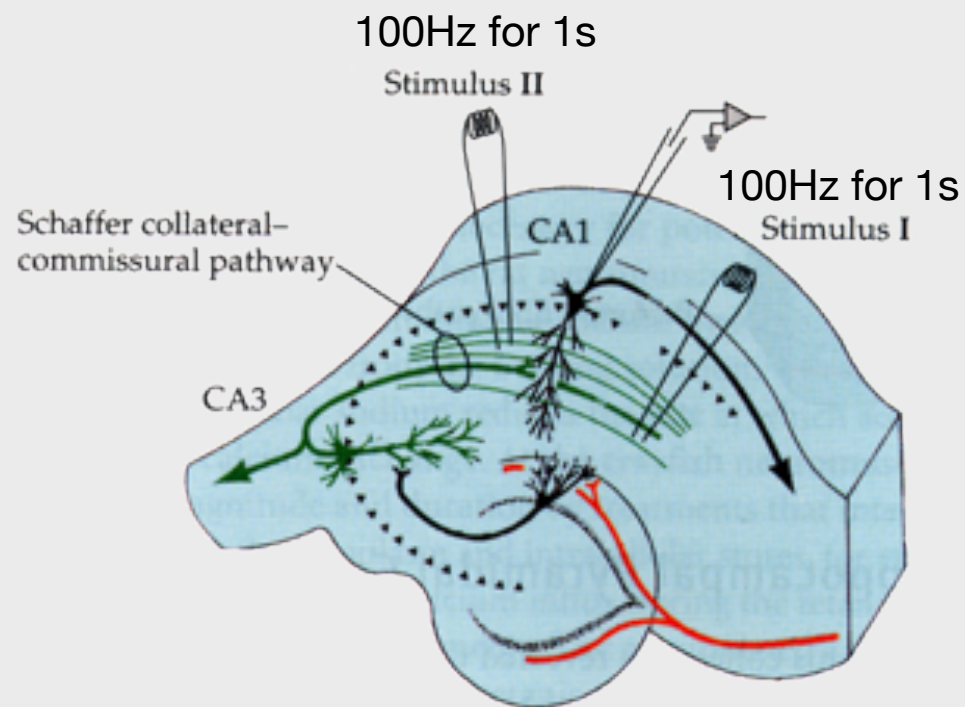
long term plasticity

exemple of bidirectional homosynaptic plasticity

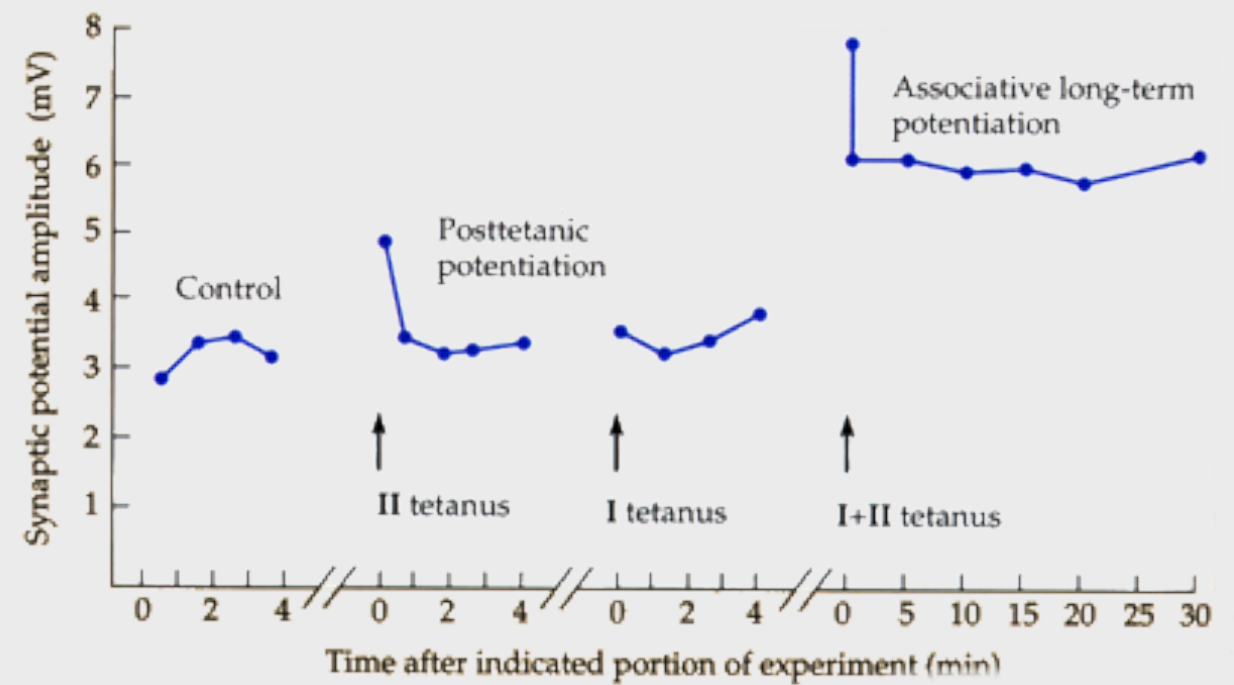


long term plasticity

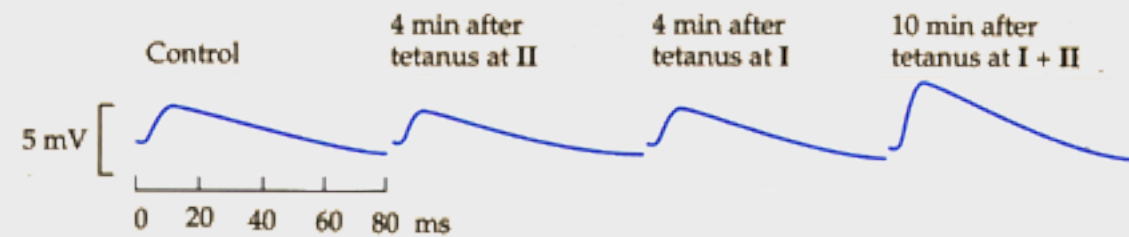
exemple of heterosynaptic plasticity



syn. II



syn. II



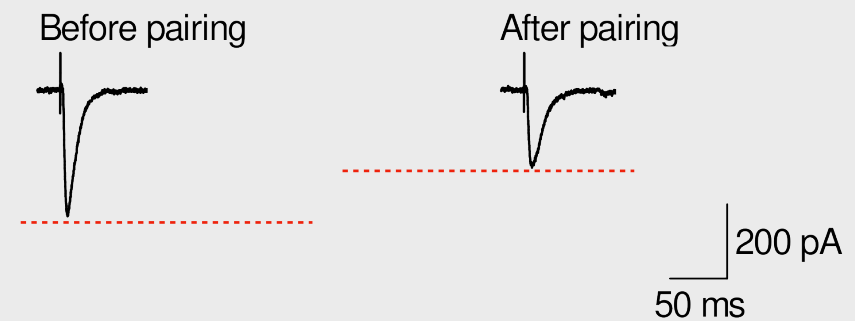
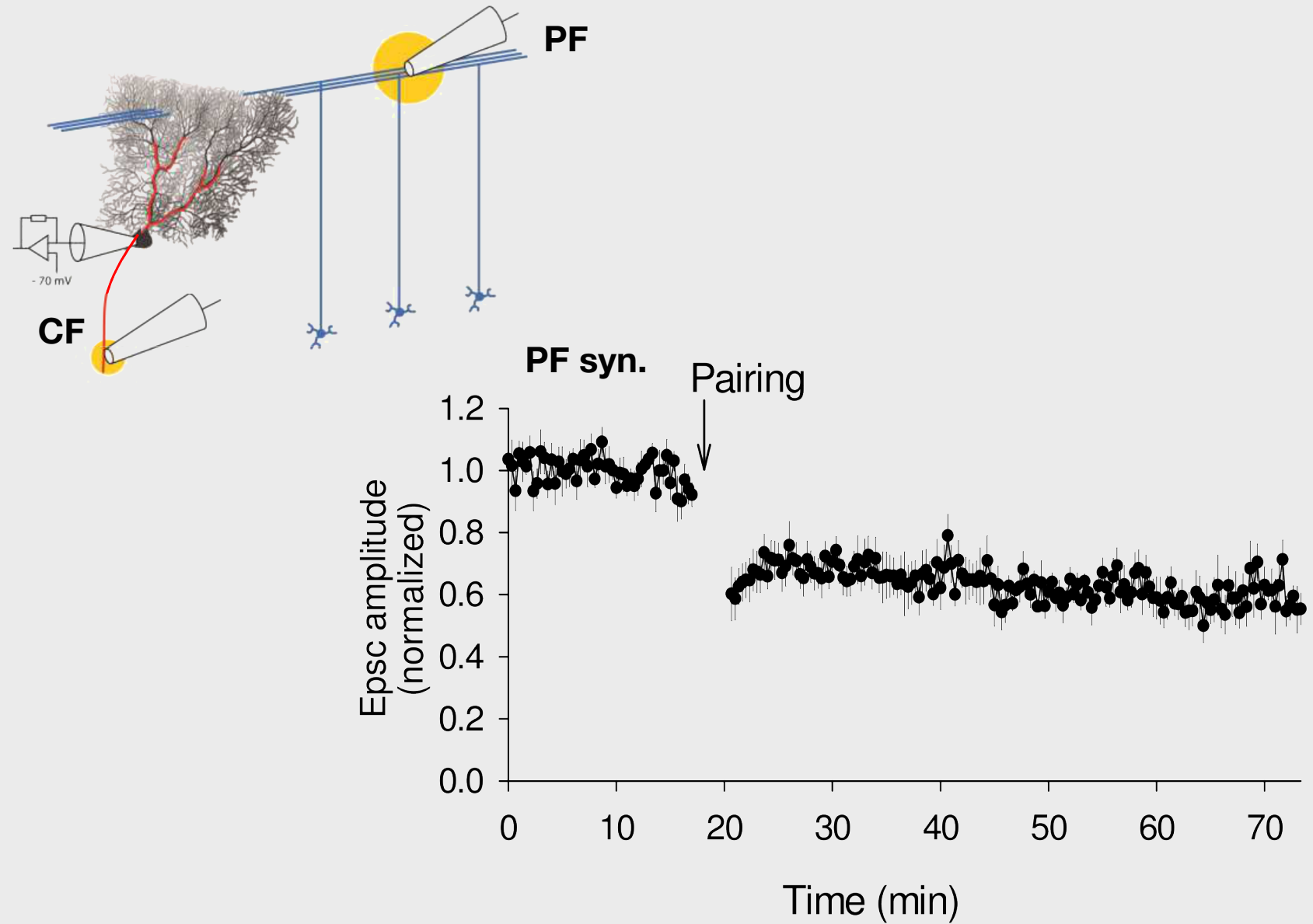
rem : homosynaptic LTP was induced at the syn. I

Barrionuevo & Brown, 1983

long term plasticity

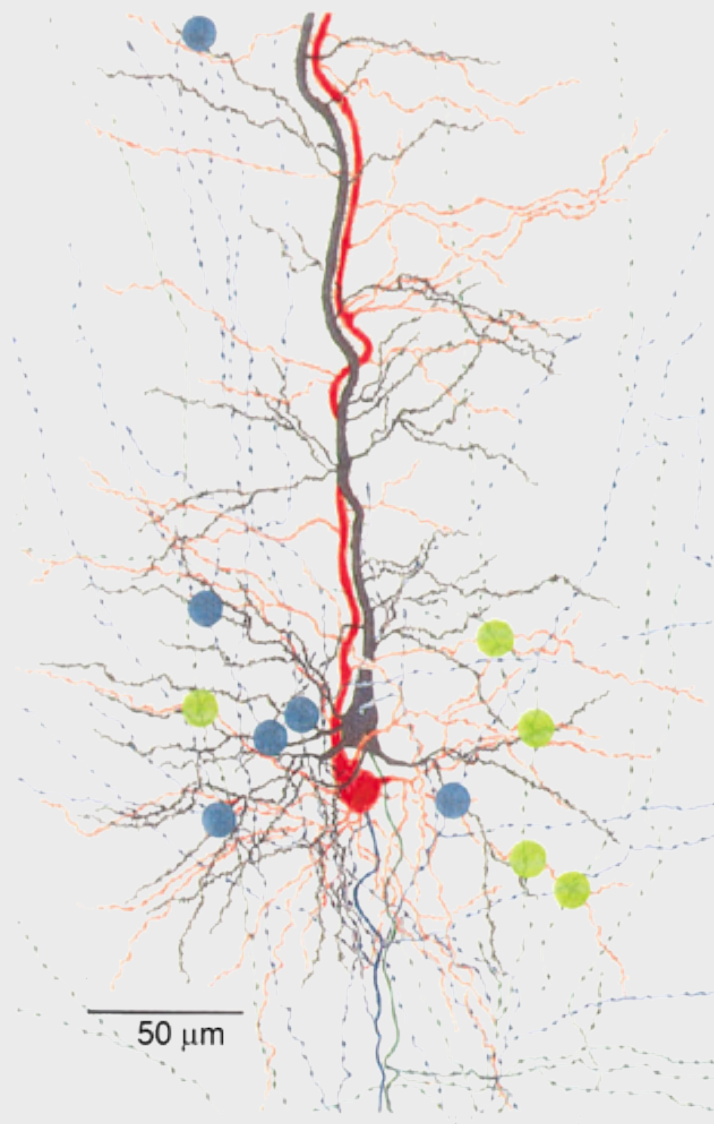
joint activation does not always induce LTP

cerebellum

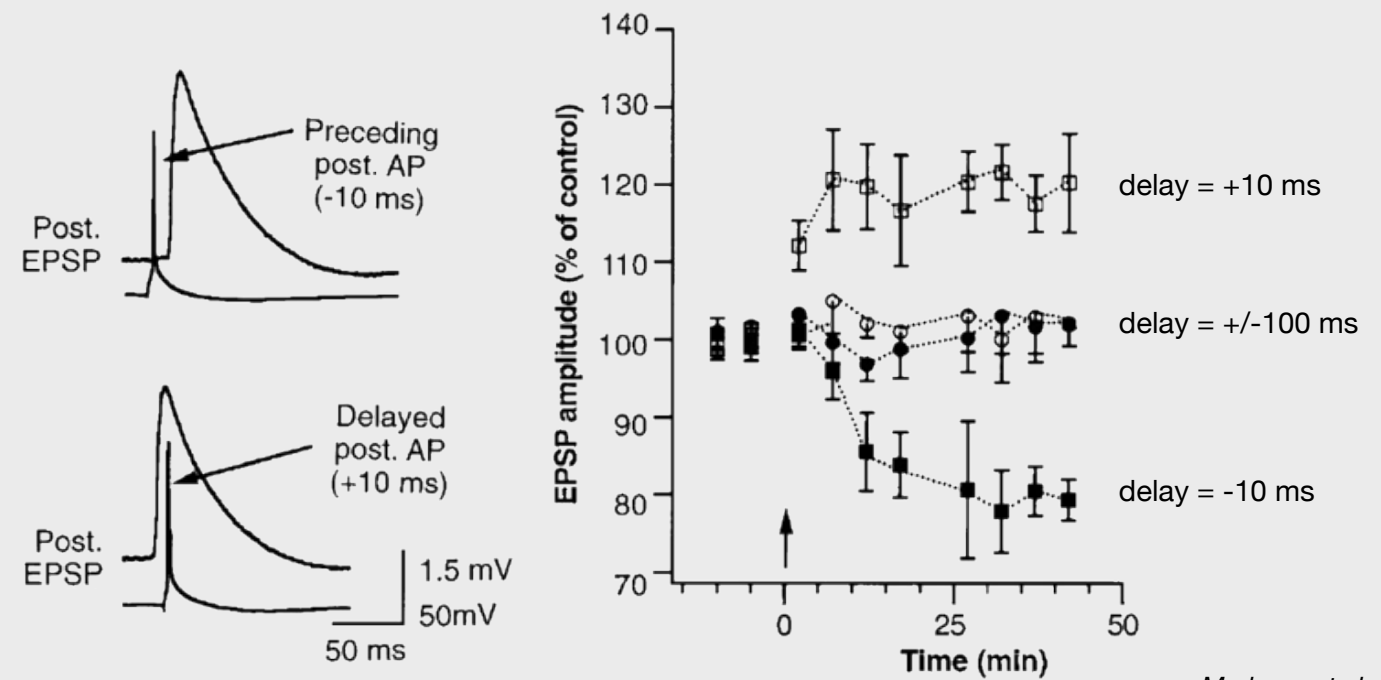


long term plasticity

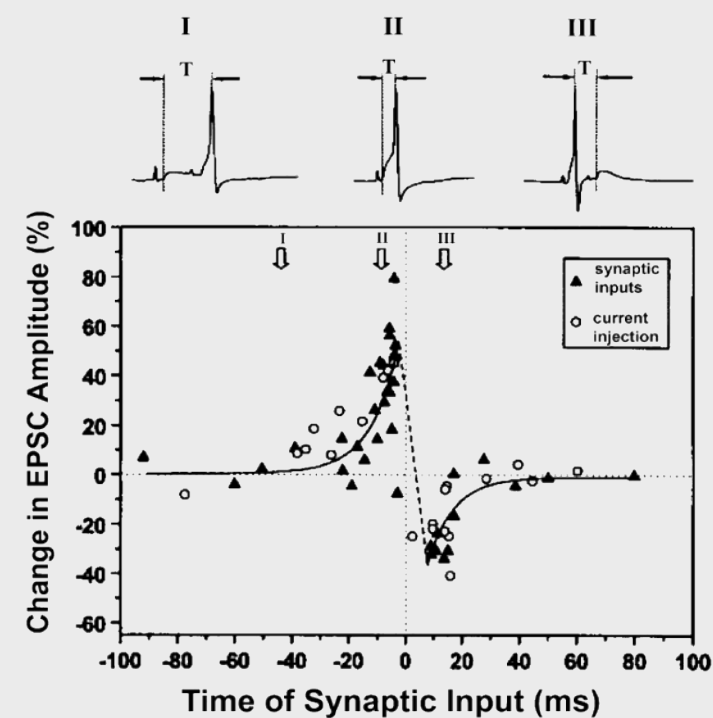
another exemple of homosynaptic plasticity : Spike Timing Dependent Plasticity



synaptically coupled pyramidal neurons
(circles : putative synaptic contacts)



Markram et al. 1997



Zhang et al. 1998

long term plasticity

triggering long term plasticity requires :

- *an intracellular messenger that links changes in V_m to intracellular pathways*
- *in many cases, coincidence detection*
- *in some cases, a retrograde messenger produced in the post-synaptic neuron inducing pre-synaptic plasticity*



?

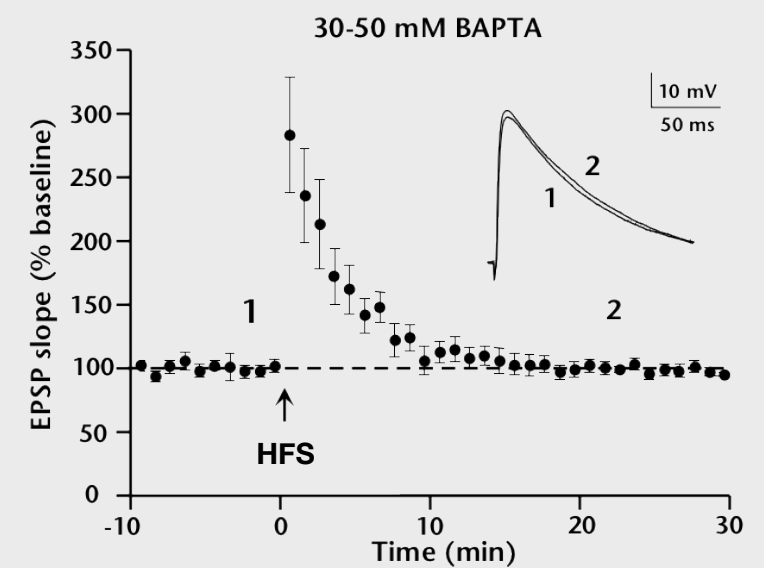
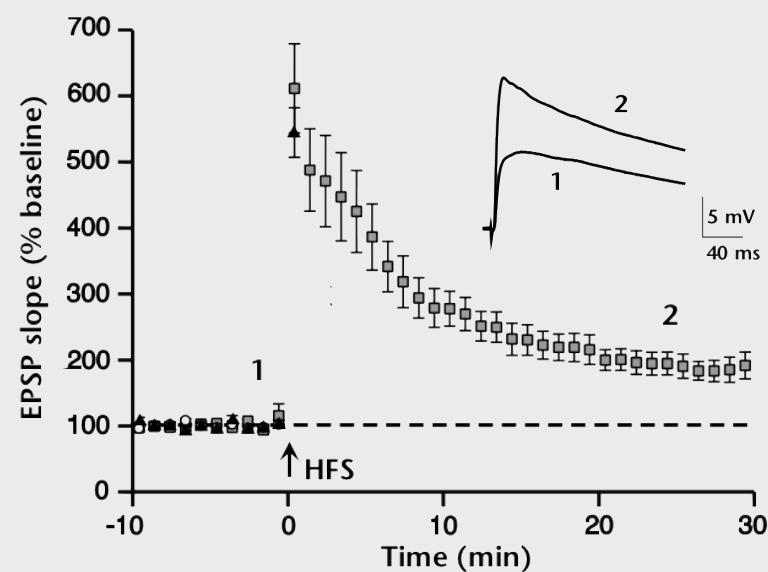
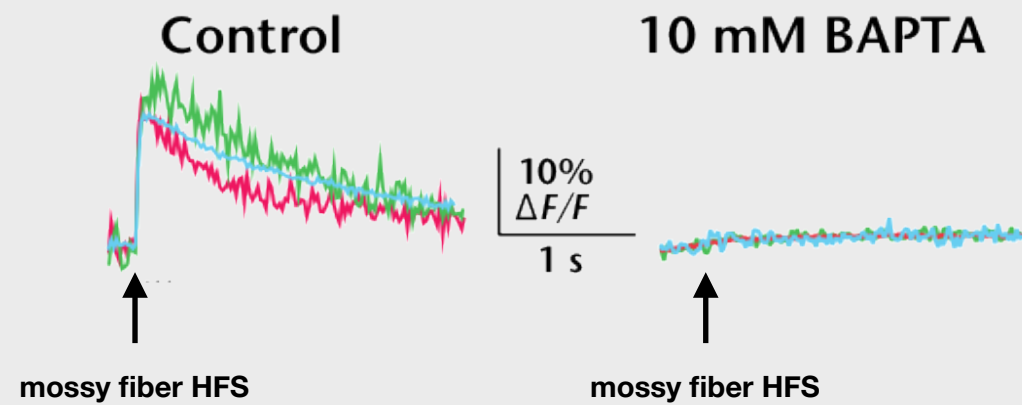
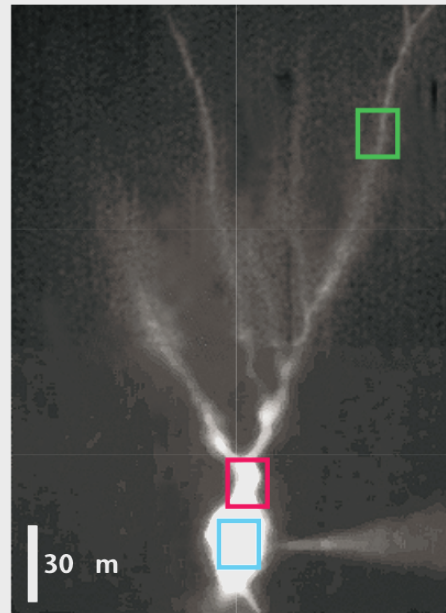
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Ca^{2+}



long term plasticity

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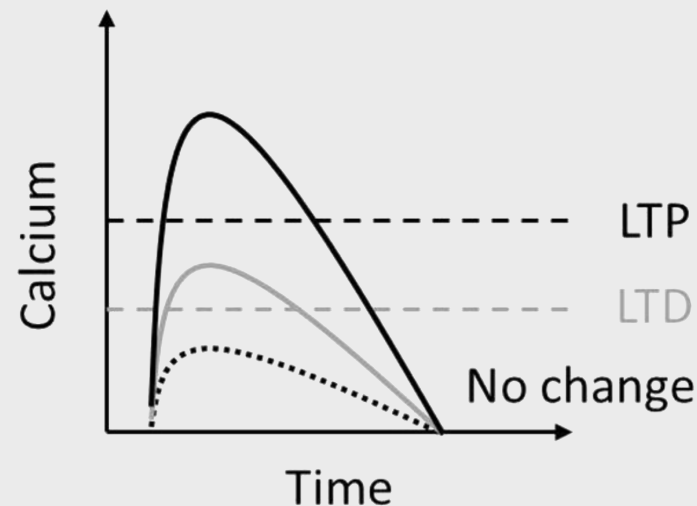


How can calcium cause synaptic strengthening (LTP) in one situation but synaptic weakening (LTD) in another ?

it probably depends on which effector is preferentially activated by the Ca^{2+} increase during the induction protocol ... for example, CAMKII initiates phosphorylation and is generally associated with LTP whereas calcineurin is a phosphatase and is generally associated with LTD.

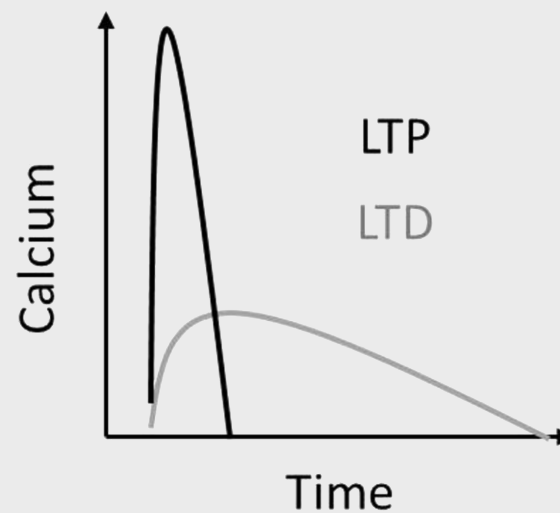
What determines which effector is activated ?

Calcium Amplitude Hypothesis



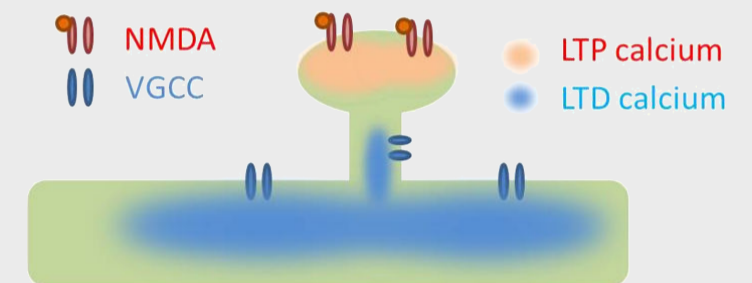
Peak calcium elevation must cross a low threshold to induce LTD, and another, higher threshold to induce LTP.

Calcium Duration Hypothesis



A low, slow calcium elevation results in LTD, while a high and fast calcium elevation results in LTP.

Calcium Location Hypothesis

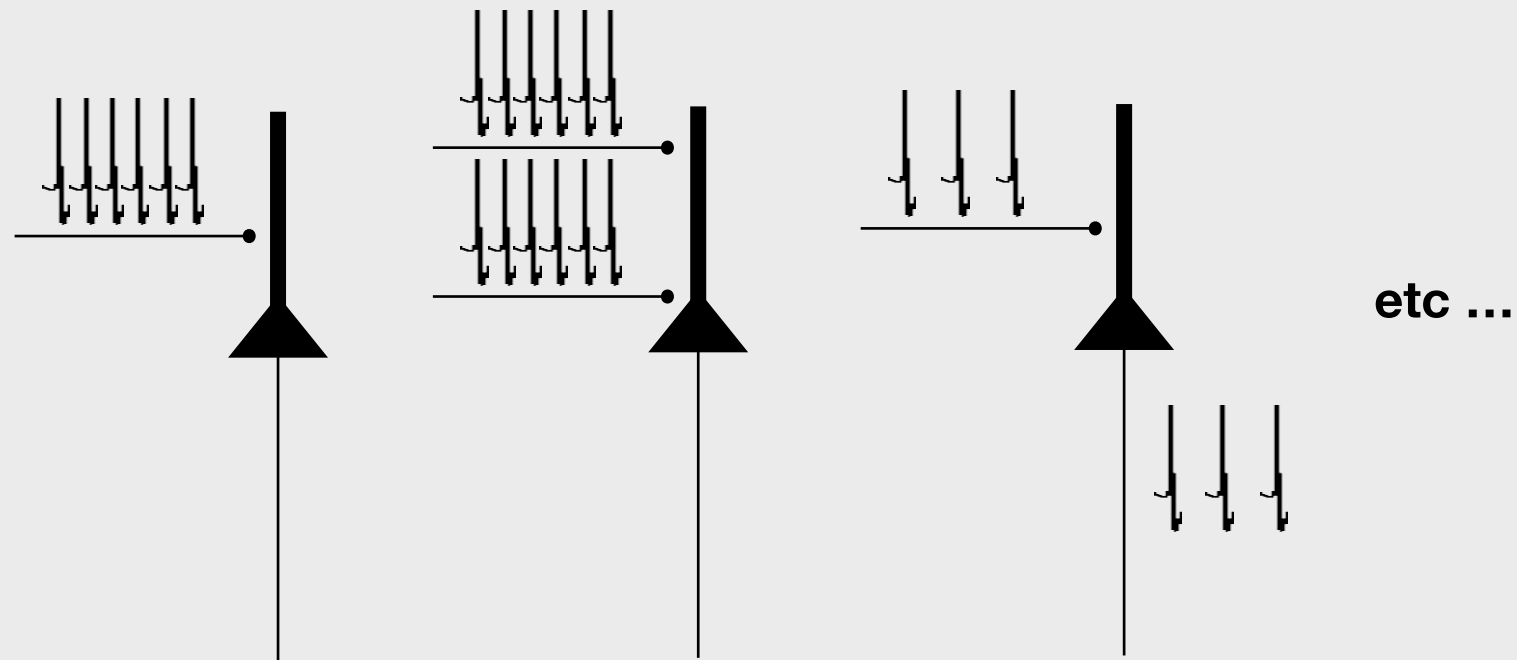


Different calcium sources may provide calcium influx in different locations creating dendrite or spine specific microdomains.

long term plasticity

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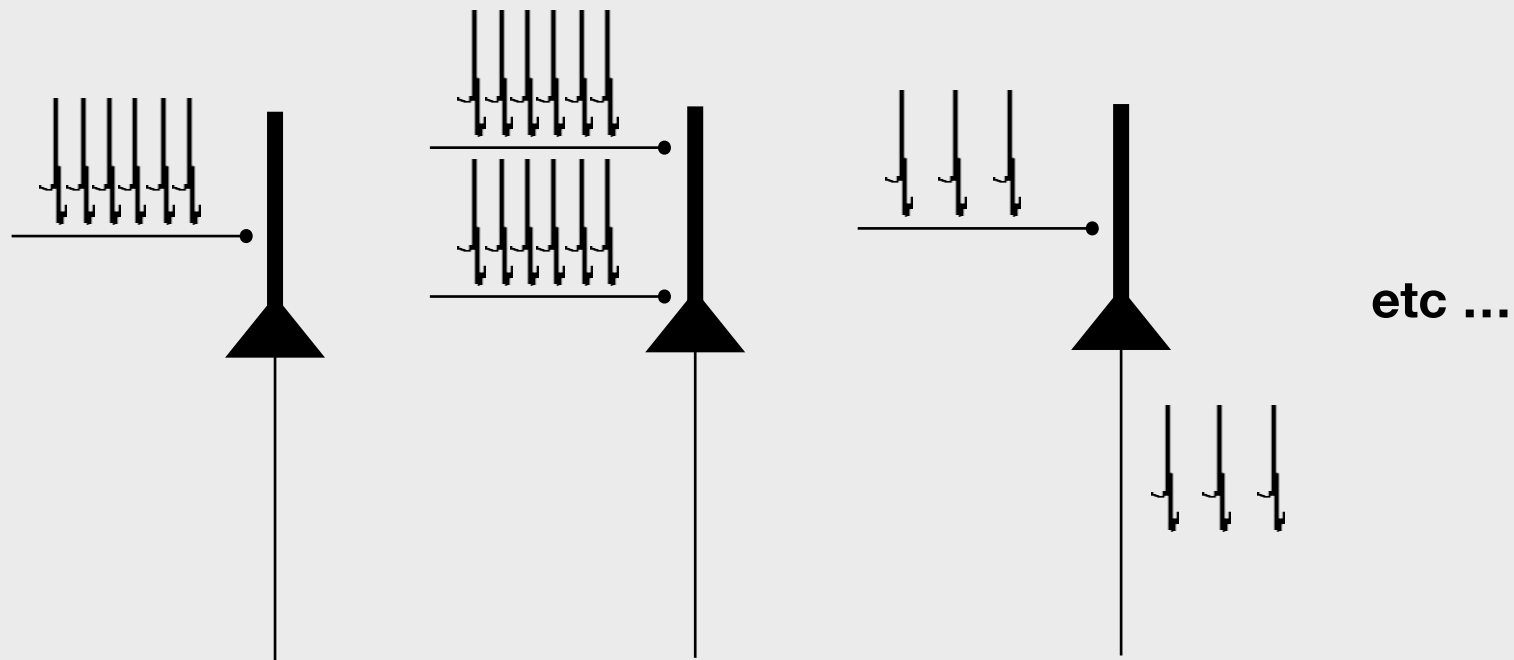
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NMDA-Rs (for excitatory synapses)



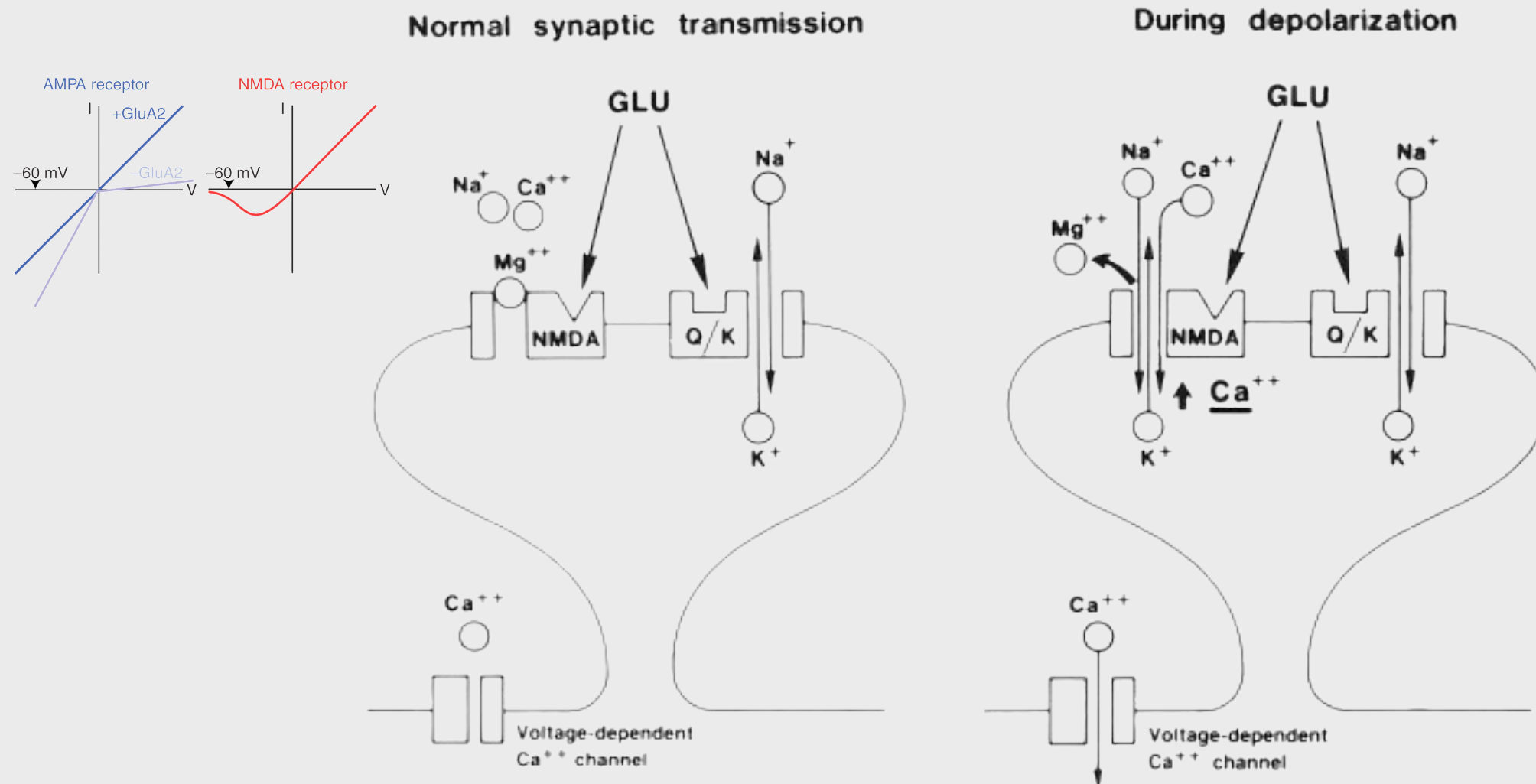
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NMDA-Rs (for excitatory synapses)



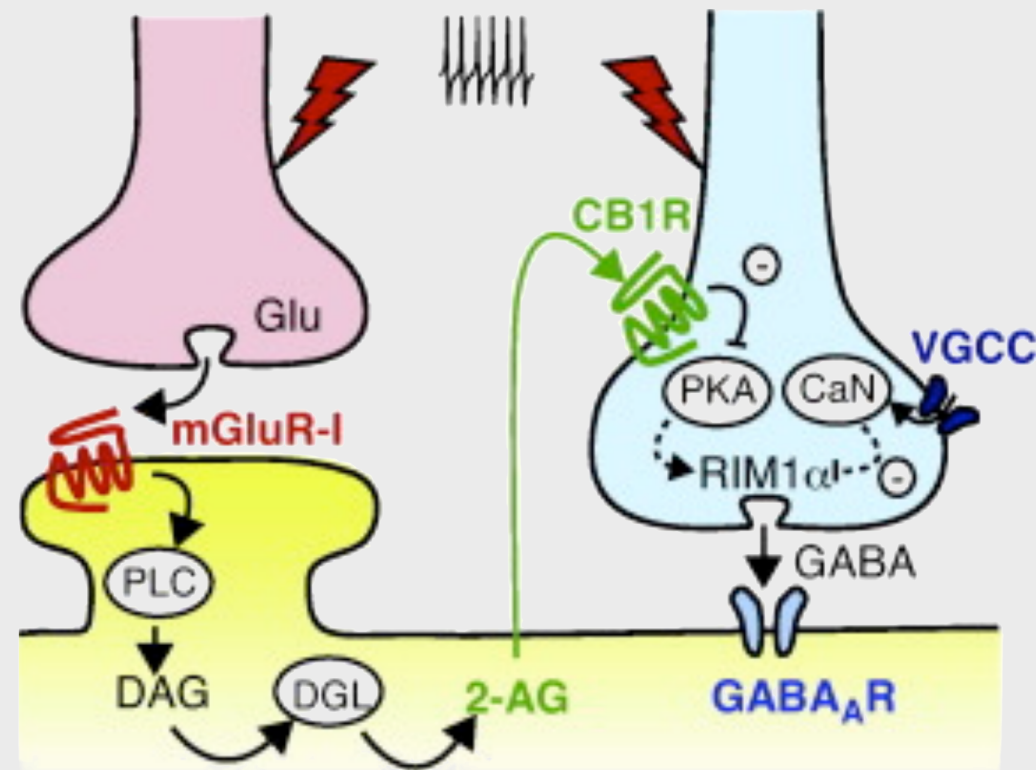
rem: there is no equivalent for inhibitory synapses and the required increase in Ca^{2+} is generally triggered by another synaptic input, heterosynaptic plasticity.

long term plasticity

triggering long term plasticity requires :

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exemple : LTD of inhibitory synapses induced by endocannabinoids



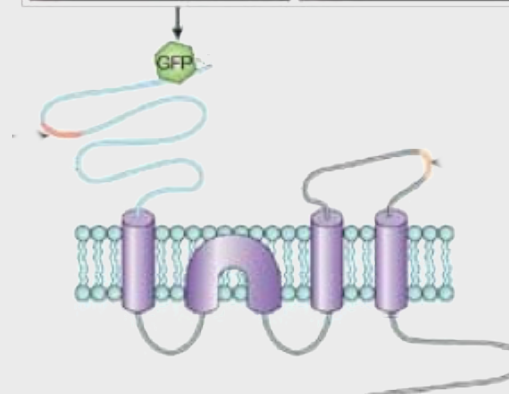
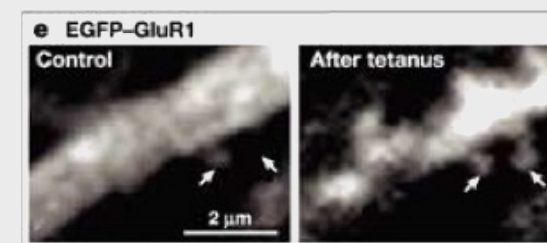
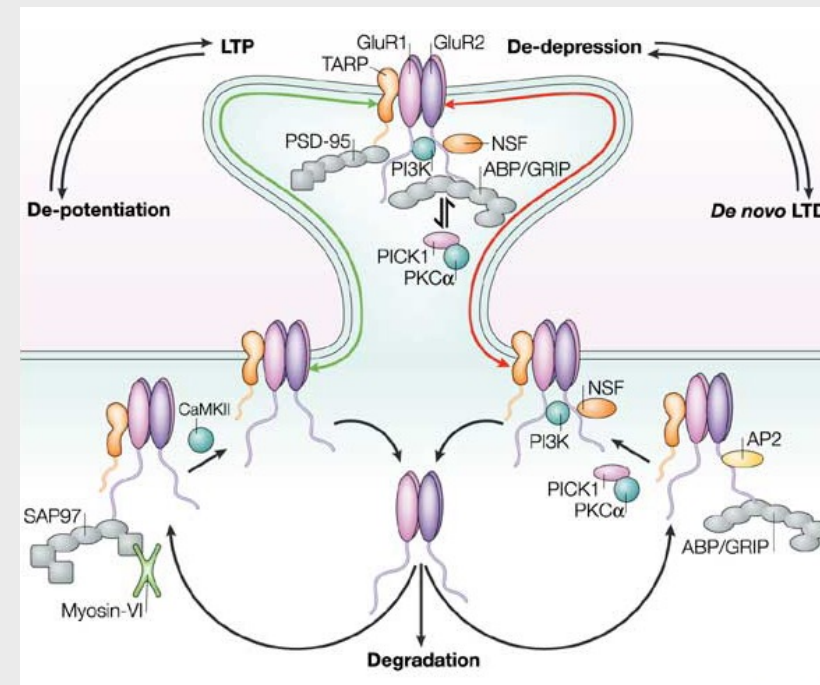
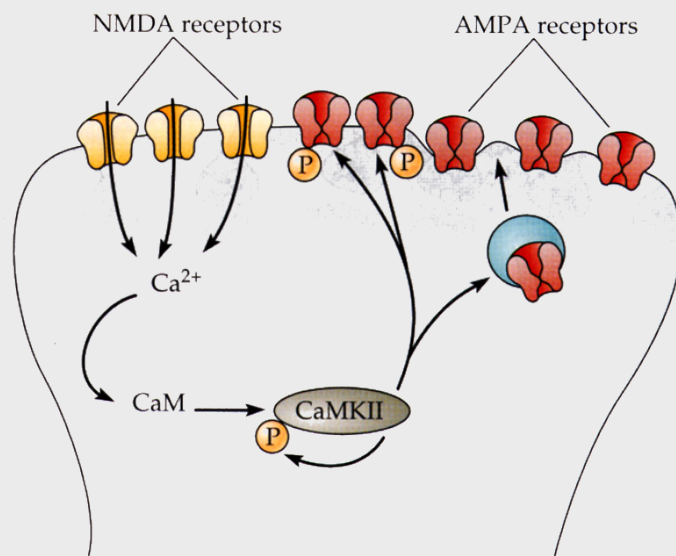
most common eCB: 2-arachidonoylglycerol (2-AG)
membrane lipid derivatives produced by Diacylglycerol lipase (DGL)

other common retrograde messenger : Nitric Oxide (NO)

long term plasticity

Long term plasticity requires multiple steps at different time scales

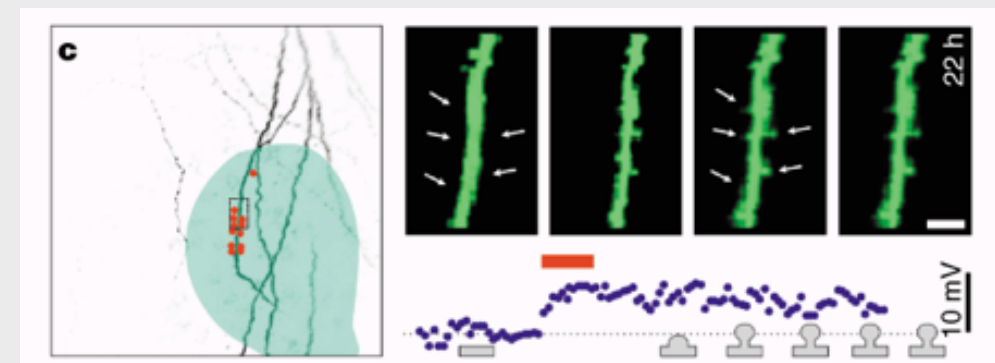
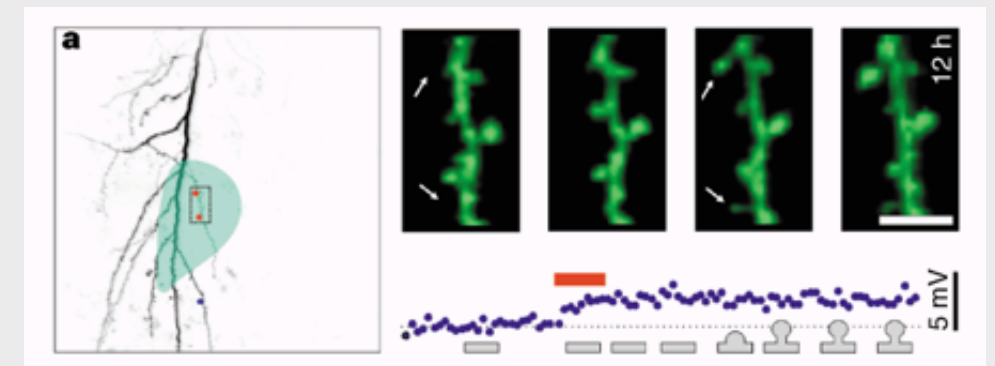
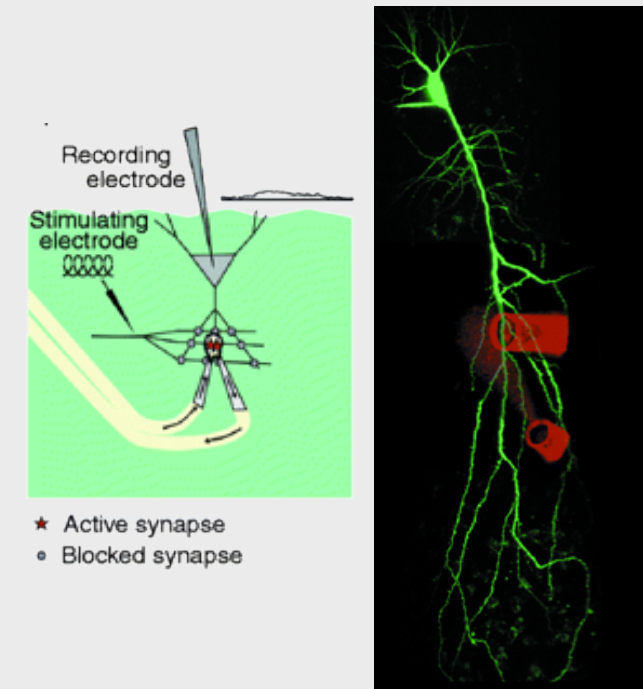
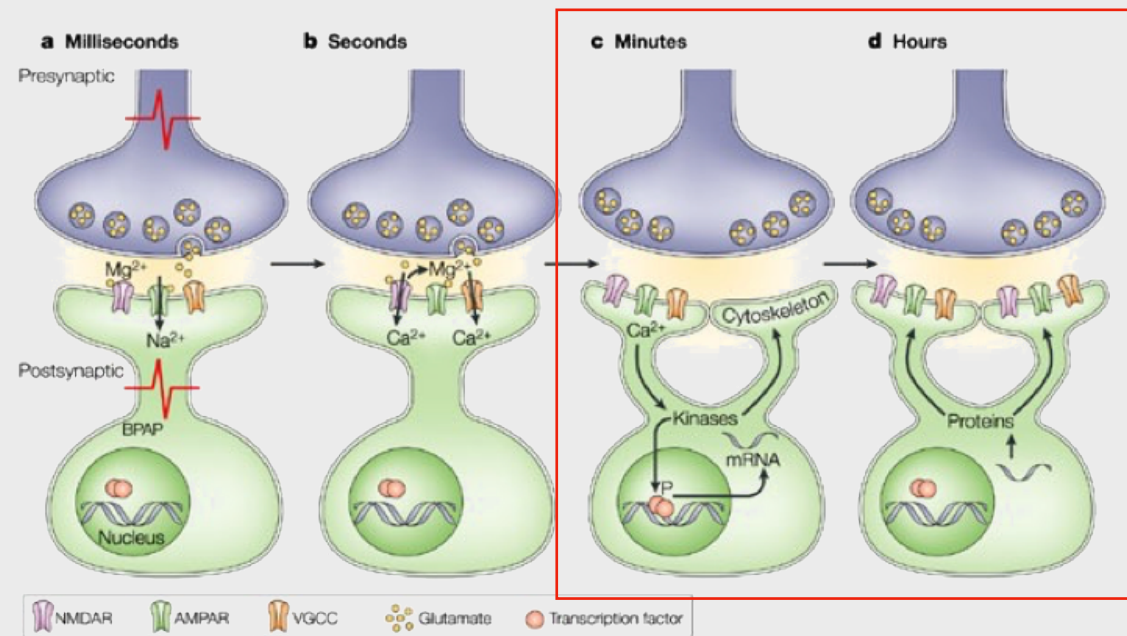
- a short initial phase lasting minutes,
- an early phase that lasts a few hours
- a late phase that lasts from several hours to weeks and potentially up to a lifetime.



long term plasticity

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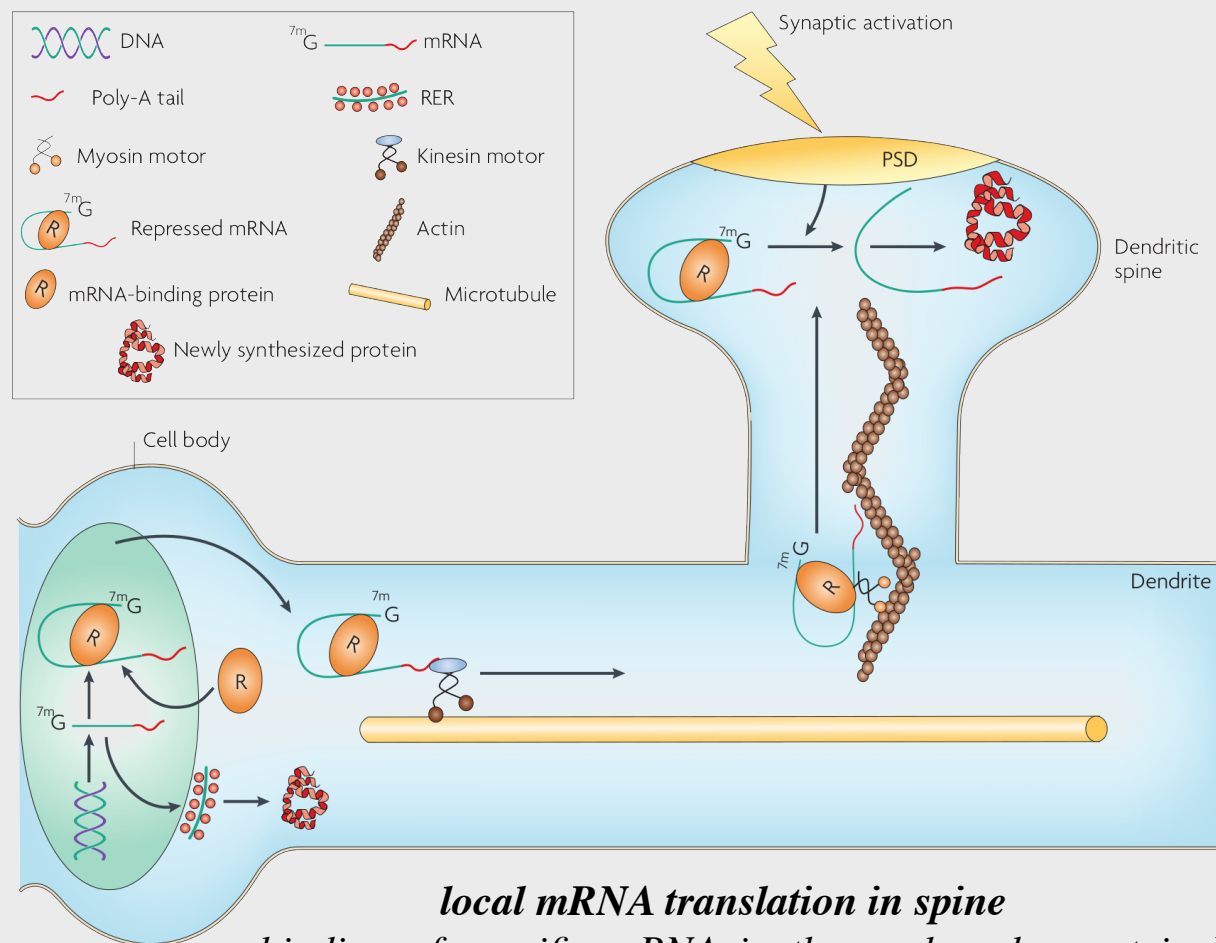


Conversely, the induction of long-term depression (LTD) induces shrinkage or loss of dendritic spines

long term plasticity

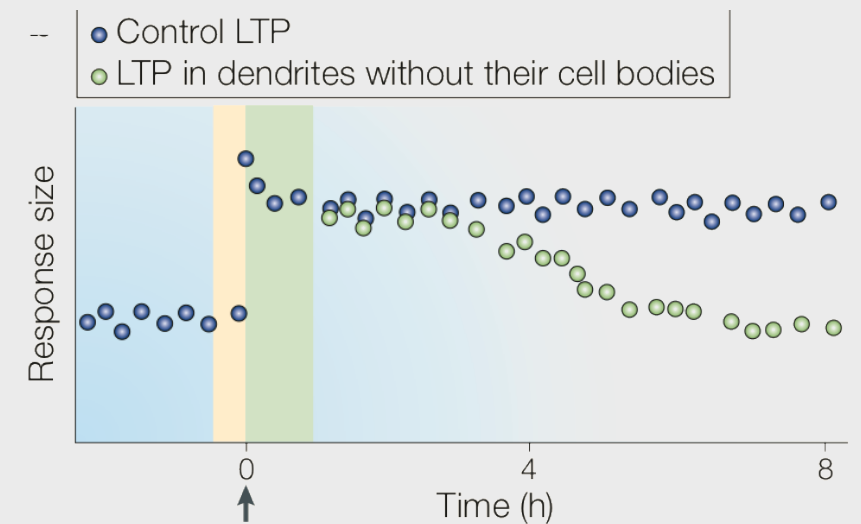
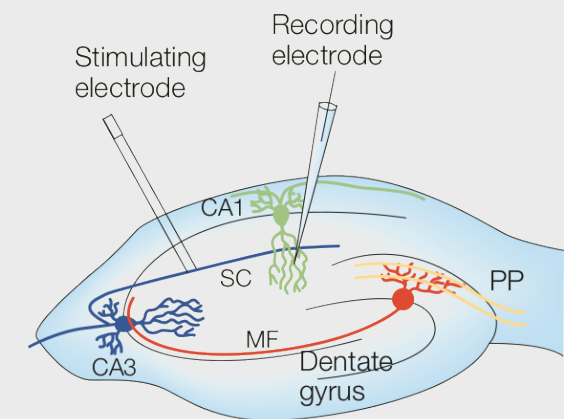
Long term plasticity requires multiple steps at different time scales

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local mRNA translation in spine

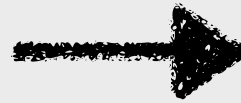
- binding of specific mRNA in the nucleus by protein R that inhibits translation
- repressed mRNAs are packaged into transport granules that are transported into the dendrite by kinesin motors on microtubules.
- following synaptic activation, translation is activated in the synaptic compartment by neutralizing R



requirement of nuclear translation ?

Homeostatic plasticity

uncompensated synaptic plasticity can lead to harmful hyper or hypo excitability of network activity



homeostatic plasticity serves to stabilize network activity.

